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Structural Portfolio

Rev 2, 2026-11-01

About

I'm a second-year Civil Engineering student at Toronto Metropolitan University (BEng, expected 2029), focused on structural engineering. I've worked on site with a design-build firm, tying rebar, building formwork and placing concrete, and as a project coordinator on a mixed-use development of two multi-storey buildings with about 182 residential and seniors-living units, coordinating five consulting disciplines (structural, civil, geotechnical, planning, landscape).

Outside class I build verification packages: I check a structure by hand, in a spreadsheet and in analysis software, chase down every difference between them, and issue drawings and models under revision control. My first package has been revised through two rounds of P.Eng. review. Tools: Robot Structural Analysis, Mathcad Prime, Excel, AutoCAD and Revit (Autodesk Certified User).

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All projects are self-directed student exercises and are not for construction.

Project 1 – Maintenance Platform Verification

Self-directed project | October 17–28, 2026 | Reviewed by S. Al-Zubaidi, P.Eng. (comments 2026-10-21 and 2026-10-28) | Revised through two rounds of P.Eng. review; next revision Dec 2026

Tools: Mathcad Prime, Excel, Robot Structural Analysis, AutoCAD, Revit

3	independent verification methods (Mathcad, Excel, Robot)
0.2%	brace and post forces, hand vs Robot (lateral load cases)
6.0 × 3.0 m	steel platform, top of steel 3,000 mm
1,298 kg	steel take-off (CISC mass per metre, members only)

Description

A 6.0 m × 3.0 m steel maintenance platform (W310 beams, W150 joists, HSS posts, L-angle X-bracing) was verified under dead, live and equipment loads. Mathcad hand calculations, an Excel calculation sheet and a Robot frame model agreed on all gravity forces within 2%. Drawn in AutoCAD and modelled in Revit, with an Excel material take-off and a review register. Revision R1 models the bracing as tension-only, adds CSA S16 notional loads and NBC guard loads, and checks every member to CSA S16. The package has been revised through two rounds of P.Eng. review (S. Al-Zubaidi, P.Eng.). Student exercise, not for construction.

Key highlights

- Bracing remodelled as tension-only: the braces carry no gravity load, which corrected the earlier R0 modelling finding.
- Hand-calculated brace and post forces under CSA S16 notional loads and NBC guard loads matched Robot within 0.2%.
- Every member checked to CSA S16 in Mathcad (bending with lateral-torsional buckling, shear, compression, tension, brace end bolt); a Class 4 joist flange was identified and checked with an effective section.
- A Revit schedule check caught a wrong brace section (modelled as W6X15 instead of L64×64×6.4), corrected and re-checked.

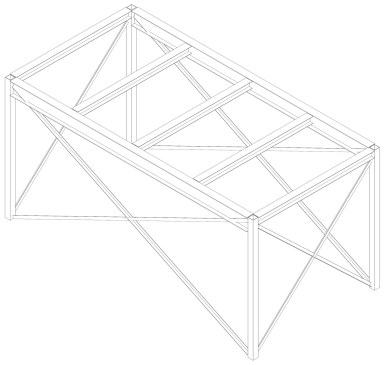


Figure 1: Revit 3D model (AS-P1-MOD-002).

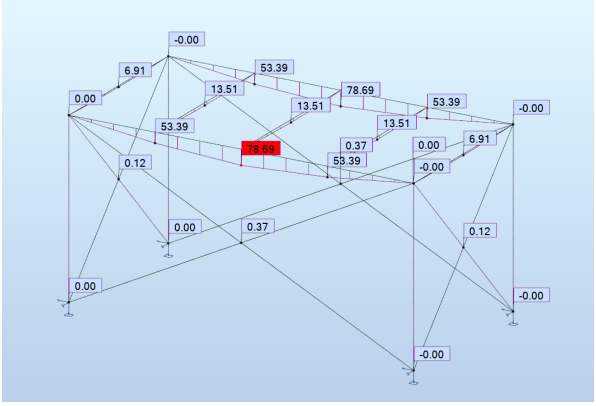
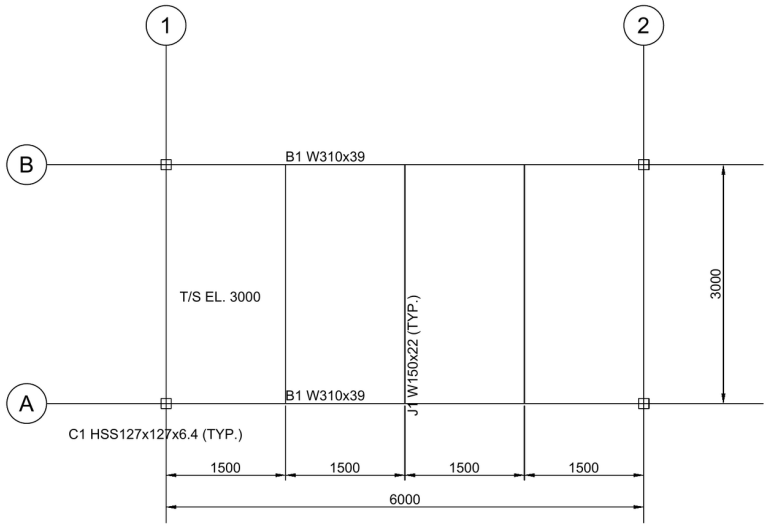


Figure 2: Robot ULS bending moments (kN·m); B1 maximum 78.69 kN·m.

P1 Steel Framing Schedule					
Mark	Type	Structural Usage	Cut Length	Count	Volume
BR1-1	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-2	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-3	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-4	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-5	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-6	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-7	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-8	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
L2-1/2X2-1/2X1/4: 8			41243 mm		0.0318 m³
J1-1	W6X15	Joist	2848 mm	1	0.0084 m³
J1-2	W6X15	Joist	2810 mm	1	0.0083 m³
J1-3	W6X15	Joist	2810 mm	1	0.0083 m³
J1-4	W6X15	Joist	2810 mm	1	0.0083 m³
J1-5	W6X15	Joist	2848 mm	1	0.0084 m³
W6X15: 5			14124 mm		0.0417 m³
B1-1	W12X26	Girder	5848 mm	1	0.0301 m³
B1-2	W12X26	Girder	5848 mm	1	0.0301 m³
W12X26: 2			11695 mm		0.0601 m³
Grand total: 15			67063 mm		0.1336 m³

P1 Column Schedule						
ss	Type	Base Level	Top Level	Count	System Length	Volume
C1-1	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-2	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-3	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-4	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
Grand total: 4						0.0355 m³

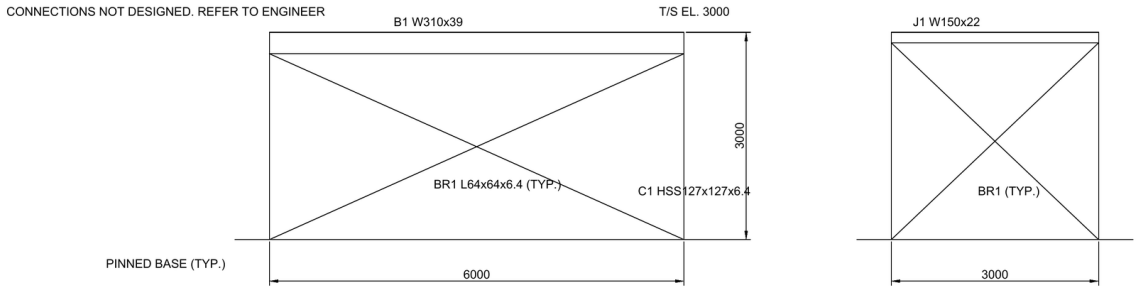
Figure 3: S-301 Revit steel framing and column schedules.



X-BRACING BR1 L64x64x6.4 ALL FOUR SIDES, BOLTED AT INTERSECTION, TENSION-ONLY. SEE S-201

GUARD LOAD 0.75 kN/m (OBC 4.1.5.14) AND S16 NOTIONAL LOADS INCLUDED IN R1 ANALYSIS. SEE AS-P1-CAL-003.

Figure 4: S-101 framing plan, 1:50 (R3).



SECTION 1 - GRID A (GRID B SIMILAR) 1:50

SECTION 2 - GRID 1 (GRID 2 SIMILAR) 1:50

GENERAL NOTES

1. STUDENT EXERCISE. NOT FOR CONSTRUCTION.
2. DESIGN BASIS: ELASTIC STRENGTH OF MATERIALS CHECK ONLY. NO CSA S16 DESIGN CHECK.
3. LOADS: GRATING 0.5 kPa + SELF-WEIGHT (DEAD); LIVE 4.8 kPa; EQUIPMENT 10 kN AT MIDSPAN OF EACH B1. ULS 1.25D + 1.5L.
4. MATERIALS: CSA G40.21 350W (ASSUMED).
5. CONNECTIONS NOT DESIGNED. BRACING AND GUARD LATERAL LOAD CHECKED AT R1 (SEE AS-P1-CAL-003/004).
6. DIMENSIONS IN mm. ELEVATIONS IN mm ABOVE GRADE.
7. SEE CALCULATION AS-P1-CAL-001 AND ROBOT MODEL AS-P1-MOD-001.
8. BR1 BOLTED AT X-BRACE INTERSECTION. BRACING TENSION-ONLY (SLENDERNESS). SEE AS-P1-MTO-001.

Figure 5: S-201 sections and general notes, 1:50 (R3).

Three-way gravity comparison (AS-P1-RPT-001 R1)

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment, M_f	kN·m	13.51	13.52	13.51	0.0%	–
J1 factored shear, V_f	kN	18.02	18.02	18.02	0.0%	–
J1 service deflection	mm	3.56	3.56	3.56	0.0%	8.33
B1 factored moment, M_f	kN·m	78.70	78.70	78.69	0.0%	–
B1 factored shear, V_f	kN	35.96	35.96	35.95	0.0%	–
B1 service deflection	mm	10.72	10.72	10.72	0.0%	16.67
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	–
C1 Euler buckling load, P_{cr}	kN	1,546	1,546	Not run	–	–

B1 deflection: Robot node displacements include post shortening; under SLS+GY the Robot B1 midspan value is 10.88 mm = 10.72 mm beam deflection + 0.16 mm post shortening (AS-P1-CAL-003). C1: the Robot reaction (46.83 kN) includes the bracing self-weight. With tension-only braces in R1, all braces are inactive under gravity and the Robot post member force is 46.55 kN, within 1.1% of the 46.03 kN hand value. Comparison criterion: 2%.

Lateral loads, hand vs Robot (AS-P1-CAL-003 R1)

Item	Case	Hand	Robot	Diff.
Brace tension, bars 22/24 (bottom/top)	ULS+GY	5.44 / 5.66 kN	5.43 / 5.66 kN	0.2% / 0.0%
C1 axial force	ULS+GY	50.40 kN	50.40 kN	0.0%
Brace tension, bars 18/20 (bottom/top)	ULS+GX	2.41 / 2.63 kN	2.41 / 2.63 kN	0.0%
C1 axial force	ULS+GX	47.63 kN	47.63 kN	0.0%
Drift at post top	SLS+GY	0.29 mm	0.30 mm	−3%
B1 lateral deflection at midspan	SLS+GY	4.67 mm	4.65 mm	+0.4%

ULS+GY = 1.25D + 1.5L + 1.5GY + 1.0NY governs the post and brace design.

CSA S16 member checks (AS-P1-CAL-004), R1 (superseded by R2)

Member	Check	Clause	Factored effect	Factored resistance	Util.
B1	Bending, Class 2, L_u = 1.5 m	13.6	M_f = 78.70 kN·m	M_r = 192.2 kN·m	0.41
B1	Shear	13.4.1.1	V_f = 35.96 kN	V_r = 373.8 kN	0.10
B1	Biaxial bending, ULS+GY	13.8.3	M_{fx} = 78.70, $M_{fy} \leq$ 5.06	M_{rx} = 192.2, M_{ry} = 42.5	**0.53**
J1	Bending, Class 4 flange, L_u = 3.0 m	13.5(c), 13.6	M_f = 13.51 kN·m	M_r = 44.3 kN·m	0.30
J1	Shear	13.4.1.1	V_f = 18.02 kN	V_r = 183.3 kN	0.10
C1	Compression, KL/r = 61.5	13.3.1	C_f = 50.40 kN	C_r = 661 kN	0.08
BR1	Tension, gross section	13.2	T_f = 5.66 kN	T_r = 241.9 kN	0.02
BR1	Net section, $0.60A_n$	12.3.3.3, 13.2	T_f = 5.66 kN	T_r = 124.7 kN	0.05
BR1 bolt	Shear, threads intercepted	13.12.1.2	V_f = 5.66 kN	V_r = 87.1 kN	0.07
BR1 bolt	Bearing on 6.35 mm leg	13.12.1.2	B_f = 5.66 kN	B_r = 137 kN	0.04

All members pass. The B1 result relies on the joists restraining its compression flange at 1.5 m (unbraced over 6.0 m, utilization 1.01); this is open and to be confirmed in the connection design.

Design basis

Members	B1 W310×39 (6.0 m, simply supported); J1 W150×22 at 1.5 m (3.0 m); C1 HSS127×127×6.4 (3.0 m, pinned, K = 1.0); BR1 L64×64×6.4 X-bracing, all four sides, tension-only, bolted at the intersection
Loads	Dead 0.5 kPa grating + self-weight; Live 4.8 kPa (NBC 2020 Table 4.1.5.3 minimum 3.6 kPa; 4.8 kPa adopted for laydown); Equipment 10 kN at midspan of each B1 (assumed); Notional 0.234 kN per post top (S16 8.4.1); Guard 0.75 kN/m on one edge at a time (NBC 2020 4.1.5.14)
Combinations	ULS 1.25D + 1.5L; ULS 1.25D + 1.5L + 1.5G + 1.0N; SLS D + L and D + L + G; Deflection L/360
Lateral support	B1 braced by J1 at 1.5 m; J1 unbraced over 3.0 m; $\omega_2 = 1.0$
Material	CSA G40.21 350W (assumed); E = 200,000 MPa; HSS Class C; brace bolt 1 M20 A325 per end (assumed)
Scope	Connections not designed apart from the brace end bolt. Student exercise, not for construction.

Full documents: Appendix A, from page 12.

Project 2 – Conveyor Gallery Truss: Member Force Verification

Self-directed project | October 2026 | Rev 0, first issue

Tools: hand calculation, Robot Structural Analysis, Excel, AutoCAD

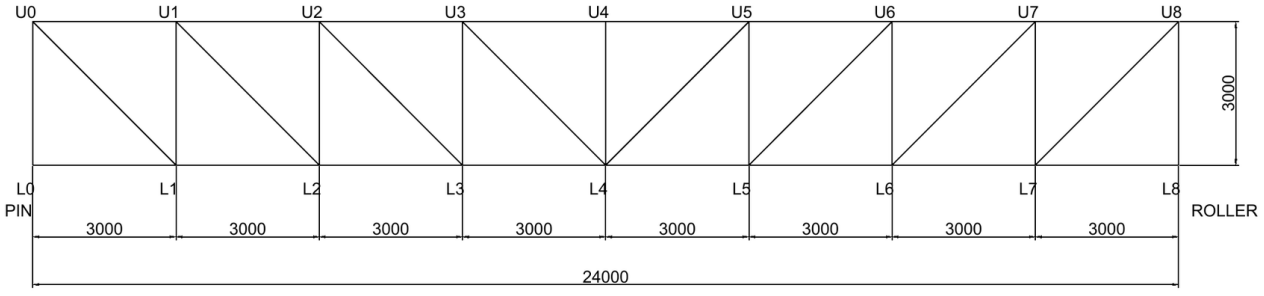
24 m	span, 8 panels × 3.0 m, 3.0 m deep
146.4 kN	maximum chord force (U3–U4, compression)
1,317 kg	steel take-off, one truss (1,449 kg with 10% connection allowance)

Description

Member-force verification of a 24 m, statically determinate Pratt truss for a conveyor gallery, with 8 panels at 3 m, 3 m deep, and pin and roller supports. Loads were worked out by hand to give reactions of 73.2 kN. Seven members were then hand-solved by the method of joints (L0, U0, L1) and a section cut through panel 4, and hand statics confirmed the Robot Structural Analysis member forces once Robot's sign convention (+ = compression) was converted. SLS deflection was checked, the elevation was drawn in AutoCAD (S-301) and a material take-off was prepared.

Key highlights

- Hand statics confirmed the Robot member forces for all 7 hand-solved members (AS-P2-CAL-002). The remaining 26 were taken from Robot and checked by symmetry.
- Governing chord force U3–U4 = 146.4 kN compression, from the section cut M/d = 439.2 / 3.0.
- Steel take-off is 1,317 kg per truss, or 1,449 kg with a 10% connection allowance.



CONVEYOR GALLERY TRUSS - ELEVATION 1:100

1. TRIAL SECTIONS: CHORDS HSS102x102x6.4; VERTICALS AND DIAGONALS L76x76x6.4. NOT DESIGN-CHECKED.
2. LOADS (ASSUMED): DEAD 2.0 kN/m, LIVE 2.4 kN/m. ULS 1.25D + 1.5L = 18.3 kN AT L1-L7.
3. MEMBER FORCES: SEE AS-P2-CAL-001 (HAND CHECK) AND AS-P2-CAL-002 (VERIFIED AGAINST ROBOT AS-P2-MOD-001).
4. CONNECTIONS NOT DESIGNED. STUDENT EXERCISE. NOT FOR CONSTRUCTION.

Figure 6: S-301 truss elevation, 1:100 (R0).

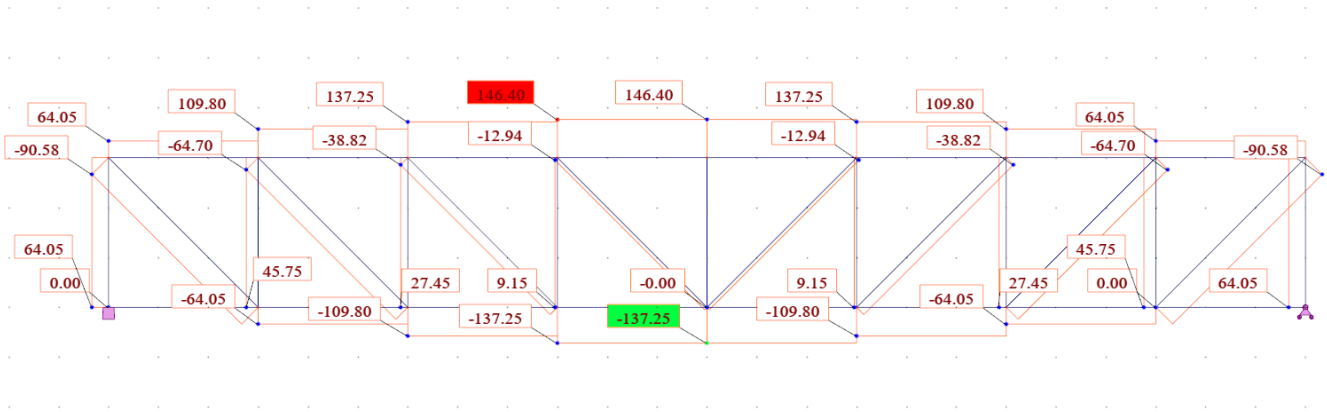


Figure 7: Robot ULS axial forces (kN); maximum 146.40 kN in U3–U4.

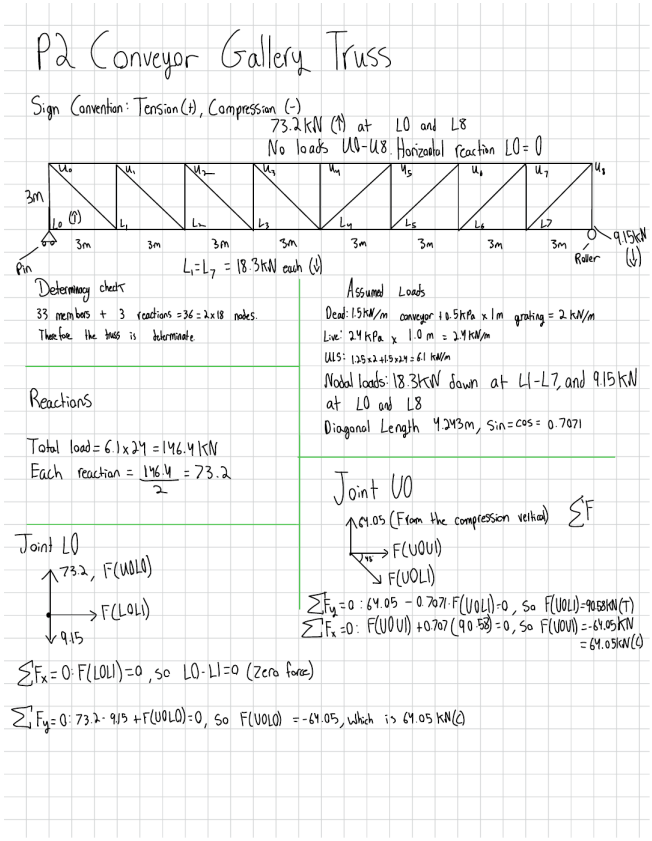
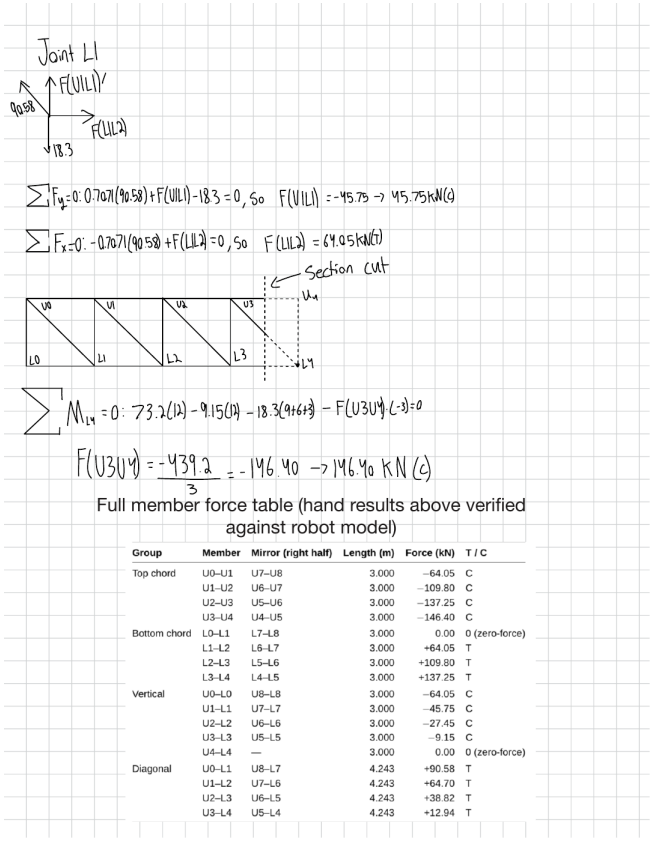


Figure 8: Hand calculation: loads, reactions, joints L0 and U0.



Full member force table (hand results above verified against robot model)

Group	Member	Mirror (right half)	Length (m)	Force (kN)	T / C
Top chord	U0–U1	U7–U8	3.000	−64.05	C
	U1–U2	U6–U7	3.000	−109.80	C
	U2–U3	U5–U6	3.000	−137.25	C
	U3–U4	U4–U5	3.000	−146.40	C
Bottom chord	L0–L1	L7–L8	3.000	0.00	0 (zero-force)
	L1–L2	L6–L7	3.000	+64.05	T
	L2–L3	L5–L6	3.000	+109.80	T
	L3–L4	L4–L5	3.000	+137.25	T
Vertical	U0–L0	U8–L8	3.000	−64.05	C
	U1–L1	U7–L7	3.000	−45.75	C
	U2–L2	U6–L6	3.000	−27.45	C
	U3–L3	U5–L5	3.000	−9.15	C
	U4–L4	—	3.000	0.00	0 (zero-force)
Diagonal	U0–L1	U8–L7	4.243	+90.58	T
	U1–L2	U7–L6	4.243	+64.70	T
	U2–L3	U6–L5	4.243	+38.82	T
	U3–L4	U5–L4	4.243	+12.94	T

Figure 9: Hand calculation: joint L1 and section cut through panel 4.

Hand-solved members vs Robot (AS-P2-CAL-002 R0)

Member	Method	Hand (kN)	Robot (kN)	Difference	T / C
L0–L1	Joint L0	0.00	0.00	0.0%	Zero force
U0–L0	Joint L0	64.05	64.05	0.0%	C
U0–L1	Joint U0	90.58	90.58	0.0%	T
U0–U1	Joint U0	64.05	64.05	0.0%	C
U1–L1	Joint L1	45.75	45.75	0.0%	C
L1–L2	Joint L1	64.05	64.05	0.0%	T
U3–U4	Section cut, panel 4	146.40	146.40	0.0%	C

Reactions: 73.20 kN each by hand and in Robot (sum 146.40 kN). The other 26 members are taken from Robot and checked by symmetry and equilibrium at joint U4; they were not solved individually by hand.

Design basis

Geometry	Pratt truss, 24.0 m span, 8 panels × 3.0 m, 3.0 m deep; pin at L0, roller at L8; 33 members, statically determinate
Trial sections	Chords HSS102×102×6.4; verticals and diagonals L76×76×6.4; not design-checked
Loads	Dead 2.0 kN/m (conveyor 1.5 kN/m + grating 0.5 kPa × 1.0 m); Live 2.4 kN/m (2.4 kPa × 1.0 m); all assumed
Combination	ULS 1.25D + 1.5L = 6.1 kN/m, giving 18.3 kN at L1–L7 and 9.15 kN at L0 and L8; self-weight excluded
Scope	Member forces only. Member resistances and connections not designed.

AS-P2-CAL-002 Hand vs Robot member forces, Conveyor Gallery Truss (ULS 1.25D + 1.5L, self-weight excluded)

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction. Robot FX sign: + = compression; reversed below to match hand convention (+ = tension).

Bar	Member	Hand (kN)	Robot FX (kN)	Robot, hand sign (kN)	Difference (kN)	% diff	Status	Hand source
1	L0–L1	0.00	0.00	0.00	0.00	0.0%	PASS	Joints L0/U0/L1
2	L1–L2	64.05	-64.05	64.05	0.00	0.0%	PASS	Joints L0/U0/L1
3	L2–L3	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
4	L3–L4	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
5	L4–L5	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
6	L5–L6	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
7	L6–L7	64.05	-64.05	64.05	0.00	0.0%	PASS	Robot (symmetry check)
8	L7–L8	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
9	U0–U1	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
10	U1–U2	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
11	U2–U3	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
12	U3–U4	-146.40	146.40	-146.40	0.00	0.0%	PASS	Section cut, panel 4
13	U4–U5	-146.40	146.40	-146.40	0.00	0.0%	PASS	Robot (symmetry check)
14	U5–U6	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
15	U6–U7	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
16	U7–U8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
17	U0–L0	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
18	U1–L1	-45.75	45.75	-45.75	0.00	0.0%	PASS	Joints L0/U0/L1
19	U2–L2	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
20	U3–L3	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
21	U4–L4	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
22	U5–L5	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
23	U6–L6	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
24	U7–L7	-45.75	45.75	-45.75	0.00	0.0%	PASS	Robot (symmetry check)
25	U8–L8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
26	U0–L1	90.58	-90.58	90.58	0.00	0.0%	PASS	Joints L0/U0/L1
27	U1–L2	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
28	U2–L3	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
29	U3–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
30	U5–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
31	U6–L5	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
32	U7–L6	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
33	U8–L7	90.58	-90.58	90.58	0.00	0.0%	PASS	Robot (symmetry check)

Reactions: hand 73.20 kN each; Robot 73.20 kN each (sum 146.40 kN). Max force 146.40 kN = M/d check.

Members with hand source 'Robot (symmetry check)' were not individually hand-solved; they satisfy symmetry and joint U4 equilibrium.

Figure 10: AS-P2-CAL-002 hand vs Robot comparison, all 33 members.

Full documents: Appendix B, from page 41.

Project 3 – Literature Review: Maintenance Access in Hydropower Plants

Self-directed literature review | October 2026 | AS-P3-RPT-001 Rev 0 | Full report with references: Appendix C, page 46

6 sources reviewed (7 references cited)
3 factors compared: fall hazards, time to establish access, inspection and maintenance
5 O. Reg. 213/91 sections cited (ss. 26, 26.1, 26.3, 130, 131)
3 questions for senior engineering review

Research question

For turbine maintenance which has to be done over and over again, what are the practical differences between using a permanent modular platform and setting up temporary scaffolding each time? This review looks at the fall hazards which come with each system, the amount of time which is needed to get access set up before the work can start, and the inspection and maintenance work which comes with each system.

Summary

From the sources which were reviewed, the main benefit of a permanent platform does not seem to come from the inspection itself. It comes from the work which has to happen before and after each inspection. The scaffold has to be built, checked, and then taken down again, and this is repeated every time access is needed. At the same time, the workers who are doing this are exposed to fall hazards while the scaffold is still incomplete. The sources which were found suggest that a permanent arrangement would be safer and would also take less time to get onto. However, no comparison was found where the hours saved or the incidents avoided were actually measured, and there was also very little information on what is needed to maintain a permanent platform over its life. Because of this, the case for a permanent platform at the moment is based on the reduction in repeated scaffold work rather than on any number which can be quoted.

Table 1. Information available from the reviewed sources

Reference	Relevant information	What remains unclear
Hatch [1]	Describes the McCormick platform and the problems which come with repeated scaffolding	Labour hours, incident data and maintenance requirements
Ontario guidance [2]	Explains mandatory working-at-heights training	Does not compare access systems
O. Reg. 213/91 [3]	Gives the requirements which apply to fall protection, guardrails and scaffolding	Does not say anything about replacing scaffolding with a permanent platform
UofT EHS [4]	Gives scaffold erection and inspection guidance	Does not address permanent turbine platforms
Robson et al. [5]	Reports changes in fall-related injury claims after the Ontario training requirement	Does not measure the effect of permanent platforms
ZARGES [6]	Describes quick setup of portable access equipment at a hydropower station	Not directly comparable to McCormick's permanent system

AS-P1-RPT-001 Technical Brief

Rev 1

Maintenance Platform Verification

2026-10-21

Prepared by: Abbas Saleh Reviewed by: S. Al-Zubaidi, P.Eng. (comments received 2026-10-18; addressed in R1)

Student exercise. Not for construction.

The gravity and lateral framing of a 6.0 m × 3.0 m steel maintenance platform was verified by three independent methods, and the review comments received on R0 were addressed in this revision.

PURPOSE AND SCOPE

The purpose of this brief is to summarize the verification of a steel maintenance platform, which was carried out as a self-directed student exercise to check the member forces, stresses and deflections of the framing by three separate methods and to record the discrepancies found between them. In R0, the checks were limited to elastic strength of materials, which covered the bending stress, shear stress and deflection of the beams and joists and the axial stress and Euler buckling of the posts. In R1, the bracing was then remodelled as tension-only, the lateral loads were checked by hand (CAL-003), and the members were checked to CSA S16 (CAL-004). Apart from the brace end bolt, connections were not designed, and for this reason the work is not intended for construction.

DESIGN BASIS

Item	Basis
Geometry	6.0 m × 3.0 m plan, top of steel 3000 mm
B1 main beams	W310×39, simply supported, 6.0 m span
J1 joists	W150×22 at 1.5 m (5 No.), 3.0 m span
C1 posts	HSS127×127×6.4, 3.0 m, pinned, K = 1.0
BR1 bracing	L64×64×6.4 X-bracing, all four sides, tension-only, bolted at the intersection
Dead load	0.5 kPa grating + self-weight
Live load	4.8 kPa adopted. The minimum for equipment areas and service rooms is 3.6 kPa (NBC 2020, Div. B, Table 4.1.5.3, as adopted by O. Reg. 163/24), and 4.8 kPa was selected to allow for components and tools laid down on the deck during maintenance (cf. storage areas, 4.8 kPa).
Equipment load	10 kN at midspan of each B1 (assumed)
Notional loads	0.005 × 187.32 kN total ULS gravity = 0.94 kN, applied as 0.234 kN at each post top, in X and Y separately (CSA S16, Clause 8.4.1)
Guard load	0.75 kN/m horizontal, outward on one edge at a time: short edge (X, 3 m) and long edge (Y, 6 m) (NBC 2020, Div. B, Article 4.1.5.14)
Load combinations	ULS 1.25D + 1.5L; ULS with lateral 1.25D + 1.5L + 1.5G + 1.0N, with full floor live load combined with guard load (conservative, for review); SLS D + L and D + L + G
Deflection limit	L/360 under D + L adopted, which is more stringent than the L/300 recommended for floors of industrial buildings in CSA S16, Annex D. For B1, D + L gives 10.72 mm (L/560) and live load only gives 9.43 mm (L/636). Checking D + L against a live load limit is conservative.
Lateral support	B1 compression flange braced by J1 at 1.5 m (L_u = 1.5 m); J1 taken as unbraced over 3.0 m, with no restraint assumed from the grating; ω_2 = 1.0
Material	CSA G40.21 350W (assumed), E = 200 000 MPa; HSS taken as Class C (n = 1.34)
Bolts	Assumed 1 M20 A325 per brace end, threads intercepted, 22 mm hole; F_u = 825 MPa (bolt), 450 MPa (350W plate)

Section properties were taken from the CISC section database (beamdimensions.com), and loads marked as assumed were not taken from a cited source.

VERIFICATION METHOD

The framing was checked by three methods which were set up independently of one another, in order that an error in any one method would be shown as a disagreement with the other two:

1. Hand calculation in Mathcad Prime (AS-P1-CAL-001), carried out member by member from joist to beam to post.
2. Parametric spreadsheet in Excel (AS-P1-CAL-002), in which every input was given as a named cell and a CHECK tab was used to compare all three methods, a difference of 2% or less being taken as a pass.
3. 3D frame model in Autodesk Robot Structural Analysis (AS-P1-MOD-001), with member self-weight included and joist end releases set to free both bending rotations while torsion was restrained.

Additionally, the framing was drawn in AutoCAD (S-101 and S-201) and modelled in Revit (AS-P1-MOD-002, sheet S-301), with the Revit schedules being used for the material take-off and as a further check of the modelled sections. In R1, the brace and post forces and the frame displacements under the lateral load cases were also checked by hand in AS-P1-CAL-003.

RESULTS

All beam and joist results were found to agree within 0.1%, and the only difference above that was the post axial load, which is explained under Findings.

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment M_f	kN·m	13.51	13.52	13.51	0.0%	
J1 factored shear V_f	kN	18.02	18.02	18.02	0.0%	
J1 service deflection	mm	3.56	3.56	3.56	0.0%	L/360 = 8.33
B1 factored moment M_f	kN·m	78.70	78.70	78.69	0.0%	
B1 factored shear V_f	kN	35.96	35.96	35.95	0.0%	
B1 service deflection	mm	10.72	10.72	10.72	0.0%	L/360 = 16.67
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	
C1 Euler buckling load P_{cr}	kN	1546	1546	not run		

The elastic stresses obtained by hand were 143 MPa bending and 20 MPa shear in B1, 85 MPa bending and 20 MPa shear in J1, and 16 MPa axial in C1, with the factored post load being about 3% of its Euler load. These values are below F_y = 350 MPa, and the member resistances to CSA S16 are given under R1 revisions. The small Mathcad and Excel differences in J1 are attributed to g being taken as 9.807 and 9.81 m/s^2 respectively.

R1 REVISIONS

Tension-only bracing. The X-bracing was modelled as tension-only in MOD-001 R1. Under gravity load all of the braces are placed in compression, which makes them inactive, and because of this the full gravity load is carried by the posts. The C1 member force from Robot was then 46.55 kN at the post base, which is within 1.1% of the 46.03 kN hand value. The support reaction of 46.83 kN is higher, since the brace self-weight is lumped at the base nodes.

Lateral loads. The notional loads and the guard load were applied as separate lateral load cases in X and Y. Since the braces act in tension only, one brace in each X is active for a given load direction. The brace and post forces were then checked by hand in CAL-003, and these agreed with Robot within 0.2%.

Item	Case	Hand	Robot	Diff.
Brace tension, bars 22/24 (bottom/top)	ULS+GY	5.44 / 5.66 kN	5.43 / 5.66 kN	0.2% / 0.0%
C1 axial force	ULS+GY	50.40 kN	50.40 kN	0.0%
Brace tension, bars 18/20	ULS+GX	2.41 / 2.63 kN	2.41 / 2.63 kN	0.0%
C1 axial force	ULS+GX	47.63 kN	47.63 kN	0.0%

ULS+GY is 1.25D + 1.5L + 1.5GY + 1.0NY, and it governs the post and brace design.

Displacements. Under SLS+GY (D + L + GY, unfactored), the drift at the post top was 0.30 mm (H/10 000), against 0.29 mm by hand. The lateral deflection of B1 at midspan was 4.65 mm (L/1290), against 4.67 mm by hand. The drift under SLS+GX was 0.15 mm.

CSA S16 member checks. The members were checked to CSA S16 in CAL-004, using the governing factored effects from the gravity and lateral load cases. All members pass, and the highest utilization is 0.53 in B1 under ULS+GY. In addition, the B1 result depends on lateral restraint from J1 (Finding 8). The weak-axis moment in B1 was taken as an upper bound, with the factored guard load of 1.125 kN/m carried by B1 alone as a simple span ($wL^2/8$), although in the frame it is shared with the joists and bracing.

Member	Check	S16 clause	Factored effect	Factored resistance	Util.
B1	Bending, Class 2, L_u = 1.5 m	13.6	M_f = 78.70 kN·m	M_r = 192.2 kN·m	0.41
B1	Shear	13.4.1.1	V_f = 35.96 kN	V_r = 373.8 kN	0.10
B1	Biaxial bending, ULS+GY	13.8.3	M_{fx} = 78.70, $M_{fy} \leq$ 5.06 kN·m	M_{rx} = 192.2, M_{ry} = 42.5 kN·m	0.53
J1	Bending, Class 4 flange, L_u = 3.0 m	13.5(c), 13.6	M_f = 13.51 kN·m	M_r = 44.3 kN·m	0.30
J1	Shear	13.4.1.1	V_f = 18.02 kN	V_r = 183.3 kN	0.10
C1	Compression, KL = 3.0 m, KL/r = 61.5	13.3.1	C_f = 50.40 kN	C_r = 661 kN	0.08
BR1	Tension, gross section yield	13.2	T_f = 5.66 kN	T_r = 241.9 kN	0.02
BR1	Net section, shear lag 0.60A_n	12.3.3.3, 13.2	T_f = 5.66 kN	T_r = 124.7 kN	0.05
BR1 bolt	Shear, threads intercepted	13.12.1.2	V_f = 5.66 kN	V_r = 87.1 kN	0.07
BR1 bolt	Bearing on 6.35 mm leg	13.12.1.2	B_f = 5.66 kN	B_r = 137 kN	0.04

In addition, the brace end connection was checked with one assumed M20 A325 bolt, and bolt shear was found to govern. The block shear, edge distances and end distances were not checked, since the connection geometry has not been fixed.

FINDINGS

The bracing was found to govern the review, since its slenderness requires a connection at the brace intersection and a tension-only arrangement. Each finding has been logged in the review register (AS-P1-MTO-001, REVIEW_REGISTER tab).

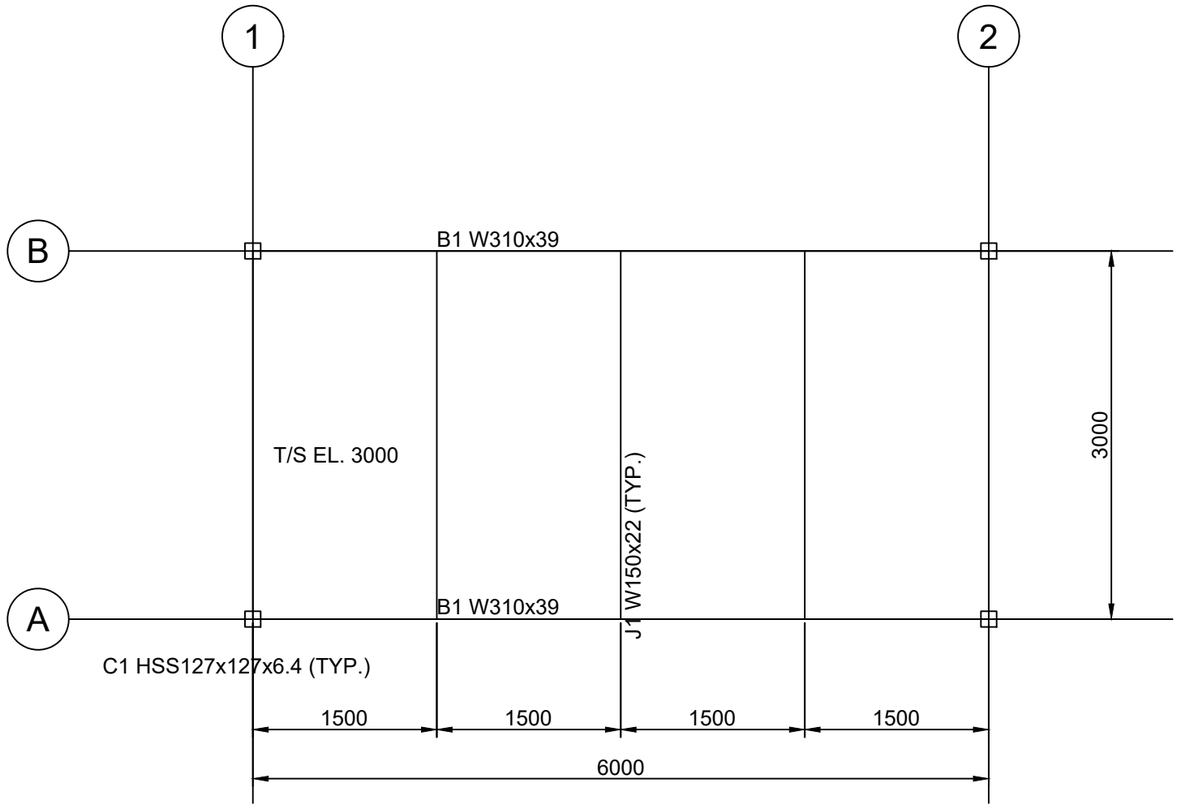
No.	Item	Finding	Action	Status
1	C1 reaction	Robot gave 46.83 kN against 46.03 kN by hand, since Robot includes the bracing self-weight (2.6 kN in total, about 0.8 kN per post at ULS).	Accepted as explained.	Closed
2	C1 member force	In R0 the Robot member force of 41.9 kN was lower than the hand value, since the bracing carried part of the gravity load.	Braces modelled as tension-only in MOD-001 R1; all braces inactive under gravity, and C1 = 46.55 kN against 46.03 kN by hand (+1.1%).	Closed
3	BR1 slenderness	With r_{min} = 12.5 mm (computed from I_{yp} and A), L/r is about 537 over the full length of the long braces, against limits of 300 in tension and 200 in compression.	Braces bolted at the intersection and treated as tension-only (S-101, S-201), and modelled as tension-only in MOD-001 R1. L/r = 268 (long) and 170 (short) at half length, below 300.	Closed
4	BR1 sag	A self-weight sag of about 24 mm (about L/280) was shown in Robot for the long braces.	Crossing bolt not modelled. In R1 the braces are truss members, so sag is not computed in Robot.	Open
5	Revit model	The schedule volume check showed the braces modelled as W6X15 (area about 2950 mm^2 against about 768 mm^2 expected).	Type corrected to the L64×64×6.4 equivalent and the schedule re-checked.	Closed
6	Material take-off	1298 kg of members was obtained from CISC mass per metre, against 1327 kg from the Revit volumes (+2.3%, imperial family geometry). An assumed 10% connection allowance gives 1428 kg in total.	Accepted, the allowance to be confirmed.	Closed
7	J1 flange class	b/2t = 11.5, which exceeds the Class 3 limit of $200/\sqrt{F_y}$ = 10.7, so the J1 flange is Class 4 in 350W.	Effective flange width used in CAL-004 (Clause 13.5(c)); utilization 0.30.	Closed
8	B1 lateral support	The B1 bending resistance assumes the compression flange is braced by J1 at 1.5 m. Unbraced over 6.0 m, the utilization would be 1.01.	Joist-to-beam connection to provide lateral restraint, to be confirmed in the connection design.	Open

CHANGES IN R1

1. Braces modelled as tension-only in MOD-001 R1 (bars 18–25). All braces are inactive under gravity, and Finding 2 is closed.
2. Notional loads (CSA S16 8.4.1) and a guard load of 0.75 kN/m (NBC 2020 4.1.5.14) added as lateral load cases. Brace and post forces and displacements checked by hand in CAL-003.
3. Live load and deflection criteria cited in the design basis (NBC 2020 Table 4.1.5.3; CSA S16 Annex D).
4. Member checks to CSA S16 added in CAL-004, including the brace end bolt (assumed 1 M20 A325; bolt shear and bearing). Other connections not designed.

DOCUMENTS ISSUED

Document	Description	Rev	Document	Description	Rev
AS-P1-CAL-001	Mathcad hand calculation	R0	AS-P1-MOD-001	Robot 3D model and result plots	R1
AS-P1-CAL-002	Excel calculation and three-way check	R0	S-101, S-201	AutoCAD plan, sections, notes	R3
AS-P1-CAL-003	Hand check of lateral loads, brace forces and displacements	R1	AS-P1-MOD-002, S-301	Revit model, schedules, sheet	R0
AS-P1-CAL-004	CSA S16 member checks	R1	AS-P1-MTO-001	MTO, slenderness, review register	R1



X-BRACING BR1 L64x64x6.4 ALL FOUR SIDES, BOLTED AT INTERSECTION, TENSION-ONLY. SEE S-201

GUARD LOAD 0.75 kN/m (OBC 4.1.5.14) AND S16 NOTIONAL LOADS INCLUDED IN R1 ANALYSIS. SEE AS-P1-CAL-003.

FRAMING PLAN 1:50

MAINTENANCE PLATFORM VERIFICATION

S-101 FRAMING PLAN

SCALE 1:50 DRAWN A. SALEH 2026-10-17

STUDENT EXERCISE. NOT FOR CONSTRUCTION

REV 0 2026-10-17 ISSUED FOR INTERNAL CHECK

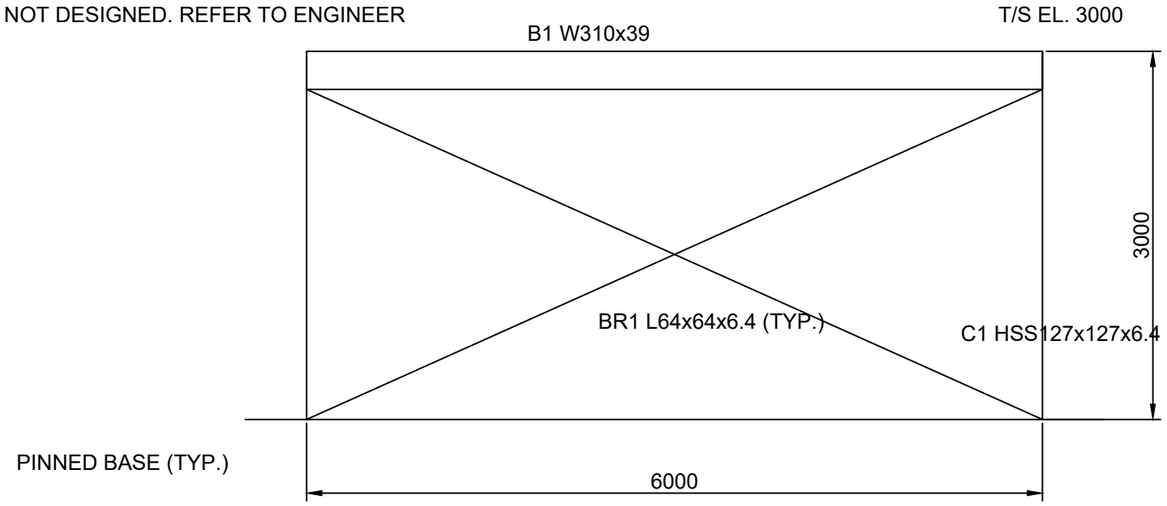
REV 1 2026-10-17 ISSUED FOR REVIEW

REV 2 2026-10-18 BR1 INTERSECTION NOTE ADDED

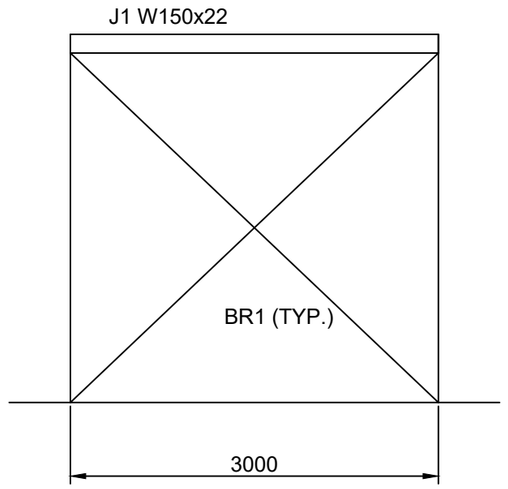
REV 3 2026-10-21 R1: TENSION-ONLY ANALYSIS, NOTIONAL AND GUARD LOADS

FRAMING PLAN 1:50

CONNECTIONS NOT DESIGNED. REFER TO ENGINEER



SECTION 1 - GRID A (GRID B SIMILAR) 1:50



SECTION 2 - GRID 1 (GRID 2 SIMILAR) 1:50

GENERAL NOTES

1. STUDENT EXERCISE. NOT FOR CONSTRUCTION.
2. DESIGN BASIS: ELASTIC STRENGTH OF MATERIALS CHECK ONLY. NO CSA S16 DESIGN CHECK.
3. LOADS: GRATING 0.5 kPa + SELF-WEIGHT (DEAD); LIVE 4.8 kPa; EQUIPMENT 10 kN AT MIDSPAN OF EACH B1. ULS 1.25D + 1.5L.
4. MATERIALS: CSA G40.21 350W (ASSUMED).
5. CONNECTIONS NOT DESIGNED. BRACING AND GUARD LATERAL LOAD CHECKED AT R1 (SEE AS-P1-CAL-003/004).
6. DIMENSIONS IN mm. ELEVATIONS IN mm ABOVE GRADE.
7. SEE CALCULATION AS-P1-CAL-001 AND ROBOT MODEL AS-P1-MOD-001.
8. BR1 BOLTED AT X-BRACE INTERSECTION. BRACING TENSION-ONLY (SLENDERNESS). SEE AS-P1-MTO-001.

MAINTENANCE PLATFORM VERIFICATION

S-201 SECTIONS AND NOTES

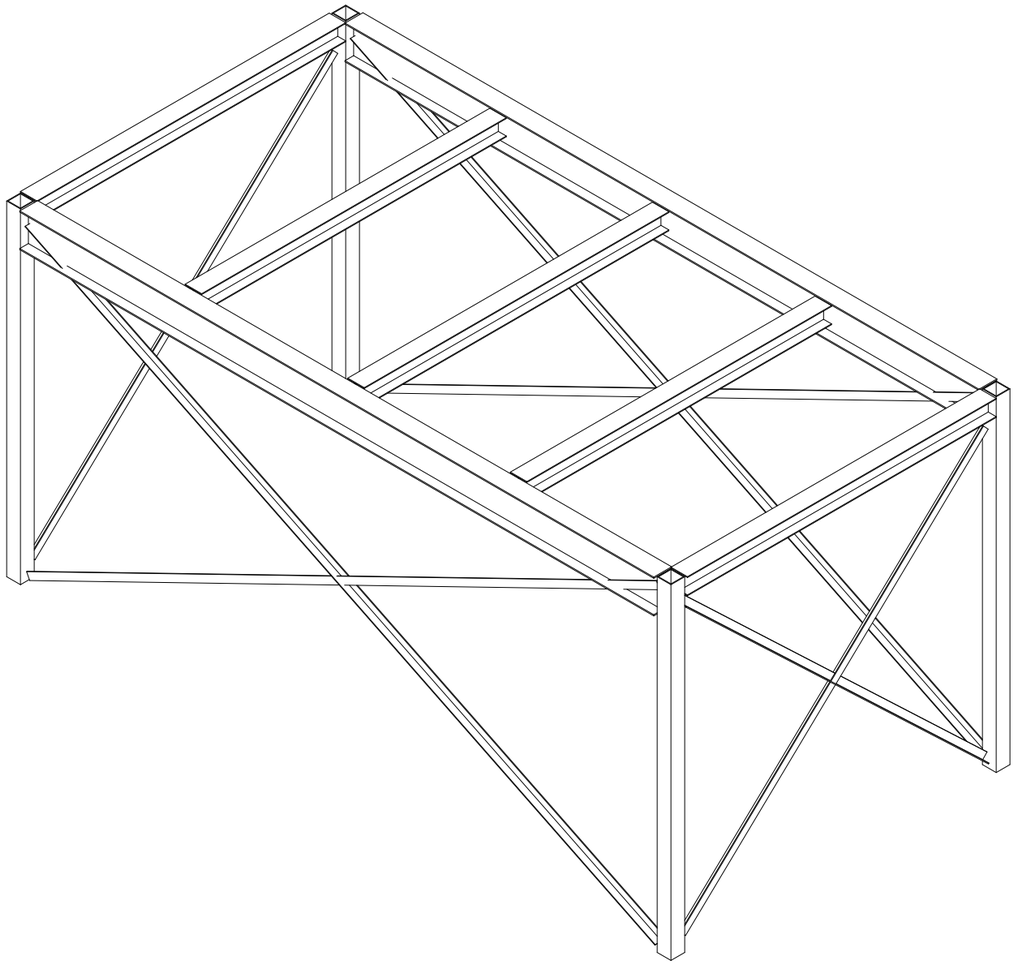
SCALE 1:50 DRAWN A. SALEH 2026-10-17

STUDENT EXERCISE. NOT FOR CONSTRUCTION

REV 0 2026-10-17 ISSUED FOR INTERNAL CHECK
REV 1 2026-10-17 ISSUED FOR REVIEW

REV 2 2026-10-18 BR1 INTERSECTION NOTE ADDED

REV 3 2026-10-21 R1: TENSION-ONLY ANALYSIS, NOTIONAL AND GUARD LOADS



STUDENT EXERCISE. NOT FOR CONSTRUCTION.

P1 Steel Framing Schedule

Mark	Type	Structural Usage	Cut Length	Count	Volume
BR1-1	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-2	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-3	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-4	L2-1/2X2-1/2X1/4	Other	6463 mm	1	0.0050 m³
BR1-5	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-6	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-7	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
BR1-8	L2-1/2X2-1/2X1/4	Other	3848 mm	1	0.0030 m³
L2-1/2X2-1/2X1/4: 8			41243 mm		0.0318 m³
J1-1	W6X15	Joist	2848 mm	1	0.0084 m³
J1-2	W6X15	Joist	2810 mm	1	0.0083 m³
J1-3	W6X15	Joist	2810 mm	1	0.0083 m³
J1-4	W6X15	Joist	2810 mm	1	0.0083 m³
J1-5	W6X15	Joist	2848 mm	1	0.0084 m³
W6X15: 5			14124 mm		0.0417 m³
B1-1	W12X26	Girder	5848 mm	1	0.0301 m³
B1-2	W12X26	Girder	5848 mm	1	0.0301 m³
W12X26: 2			11695 mm		0.0601 m³
Grand total: 15			67063 mm		0.1336 m³

P1 Column Schedule

ss	Type	Base Level	Top Level	Count	System Length	Volume
C1-1	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-2	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-3	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
C1-4	HSS5X5X1/4	GRADE	T/S	1	3000	0.0089 m³
Grand total: 4						0.0355 m³

No.	Description	Date

Maintenance Platform Verification

Steel Model and Schedules

Project number	AS-P1
Date	2026-10-18
Drawn by	A. Saleh
Checked by	

S-301

Scale

2026-10-18 1:37:16 PM

AS-P1-CAL-001 Rev 0. Maintenance Platform Hand Calculation. Abbas Saleh, Oct 17 2026. Student exercise, not for construction.

Joist J1

$LJ := 3 \cdot m$
$sJ := 1.5 \cdot m$
$qD := 0.5 \cdot kPa$
$qL := 4.8 \cdot kPa$
$mJ := 22.5 \ \frac{kg}{m}$

$E := 200000 \cdot MPa$
$Sx := 159 \cdot 10^3 \cdot mm^3$
$Ix := 12.1 \cdot 10^6 \cdot mm^4$
$d := 152 \cdot mm$
$tw := 5.8 \cdot mm$
$wsw := mJ \cdot g$
$wD := qD \cdot sJ + wsw$
$wL := qL \cdot sJ$
$wF := 1.25 \cdot wD + 1.5 \cdot wL$

$$Mf := wF \cdot \frac{LJ^2}{8}$$
$$Vf := wF \cdot \frac{LJ}{2}$$
$$\sigma := \frac{Mf}{Sx}$$
$$\tau := \frac{Vf}{(d \cdot tw)}$$
$ws := wD + wL$

$$\delta := 5 \cdot ws \cdot \frac{LJ^4}{(384 \cdot E \cdot Ix)}$$

$$\delta lim := \frac{LJ}{360}$$

$Mf = 13.515 \ kN \cdot m$
$Vf = 18.02 \ kN$
$\sigma = 85 \ MPa$
$\tau = 20.44 \ MPa$
$\delta = 3.561 \ mm$
$\delta lim = 8.333 \ mm$

Main beam B1

$LB := 6 \cdot m$

$mB := 38.8 \cdot \frac{kg}{m}$

$SxB := 549 \cdot 10^3 \cdot mm^3$

$IxB := 85.1 \cdot 10^6 \cdot mm^4$

$dB := 310 \cdot mm$

$twB := 5.8 \cdot mm$

$Peq := 10 \cdot kN$

$wswB := mB \cdot g$

$PD := wD \cdot \frac{LJ}{2}$

$PL := wL \cdot \frac{LJ}{2}$

$Pf := 1.25 \cdot PD + 1.5 \cdot PL$

$Peqf := 1.5 \cdot Peq$

$wBf := 1.25 \cdot wswB$

$RB := \frac{(3 \cdot Pf + Peqf)}{2} + wBf \cdot \frac{LB}{2}$

$MB := RB \cdot 0.5 \cdot LB - Pf \cdot 1.5 \cdot m - 0.125 \cdot wBf \cdot LB^2$

$\sigma B := \frac{MB}{SxB}$

$\tau B := \frac{RB}{(dB \cdot twB)}$

$Ps := PD + PL$

$a1 := 1.5 \cdot m$

$\delta 1 := 2 \cdot Ps \cdot a1 \cdot \frac{\langle 3 \cdot LB^2 - 4 \cdot a1^2 \rangle}{(48 \cdot E \cdot IxB)}$

$\delta 2 := (Ps + Peq) \cdot \frac{LB^3}{(48 \cdot E \cdot IxB)}$

$\delta 3 := 5 \cdot wswB \cdot \frac{LB^4}{(384 \cdot E \cdot IxB)}$

$\delta B := \delta 1 + \delta 2 + \delta 3$

$\delta Blim := \frac{LB}{360}$

$RB = 35.957\ \boldsymbol{kN}$
$MB = 78.7\ \boldsymbol{kN \cdot m}$
$\sigma B = 143.352\ \boldsymbol{MPa}$
$\tau B = 19.998\ \boldsymbol{MPa}$
$\delta B = 10.717\ \boldsymbol{mm}$
$\delta Blim = 16.667\ \boldsymbol{mm}$

Post C1

$HC := 3 \cdot \boldsymbol{m}$
$K := 1$
$AC := 2960 \cdot \boldsymbol{mm}^2$
$IC := 7.05 \cdot 10^6 \cdot \boldsymbol{mm}^4$

$mC := 23.2 \cdot \dfrac{\boldsymbol{kg}}{\boldsymbol{m}}$

$wDe := qD \cdot 0.75 \cdot \boldsymbol{m} + wsw$
$wLe := qL \cdot 0.75 \cdot \boldsymbol{m}$

$Re := \left(1.25 \cdot wDe + 1.5\ wLe\right) \cdot 0.5 \cdot LJ$
$PC := RB + Re + 1.25 \cdot mC \cdot g \cdot HC$

$Pcr := \pi^2 \cdot E \cdot \dfrac{IC}{\left(K \cdot HC\right)^2}$

$\sigma C := \dfrac{PC}{AC}$

$ratioC := \dfrac{PC}{Pcr}$

$PC = 46.027\ \boldsymbol{kN}$
$Pcr = \left(1.546 \cdot 10^3\right)\ \boldsymbol{kN}$
$\sigma C = 15.55\ \boldsymbol{MPa}$
$ratioC = 0.03$

INPUT AS-P1-INP-001 Input Sheet

GEOMETRY	Symbol	Value	Unit	Source / assumption
Main beam span (grid 1 to 2)	L_B1	6	m	Assumed scenario
Joist span (grid A to B)	L_J1	3	m	Assumed scenario
Joist spacing	s_J	1.5	m	Assumed scenario
Number of joists	n_J	5	-	At 0, 1.5, 3.0, 4.5, 6.0 m
Post height (grade to top of steel)	H_C1	3	m	Assumed scenario
Post effective length factor	K	1	-	Assumed: braced frame, pinned-pinned posts
Equipment load position on B1	x_eq	3	m	Midspan of B1

MATERIAL	Symbol	Value	Unit	Source / assumption
Yield strength	Fy	350	MPa	Assumed CSA G40.21 350W
Elastic modulus	E	200000	MPa	Standard value for structural steel
Gravity	g	9.81	m/s2	Standard

SECTION PROPERTIES (fill from CISC database)	Symbol	Value	Unit	Source / assumption
B1 W310x39: area	A	4940	mm2	beamdimensions.com, CISC W310x39
B1 W310x39: depth	d	310	mm	beamdimensions.com, CISC W310x39
B1 W310x39: web thickness	w	5.8	mm	beamdimensions.com, CISC W310x39
B1 W310x39: moment of inertia (strong axis)	Ix	85.1	x10^6 mm4	beamdimensions.com, CISC W310x39
B1 W310x39: elastic section modulus	Sx	549	x10^3 mm3	beamdimensions.com, CISC W310x39
B1 W310x39: mass	m	38.8	kg/m	beamdimensions.com, CISC W310x39
B1 self-weight (formula)	w_sw	0.380628	kN/m	Formula: mass x g / 1000
J1 W150x22: area	A	2860	mm2	beamdimensions.com, CISC W150x22
J1 W150x22: depth	d	152	mm	beamdimensions.com, CISC W150x22
J1 W150x22: web thickness	w	5.8	mm	beamdimensions.com, CISC W150x22
J1 W150x22: moment of inertia (strong axis)	Ix	12.1	x10^6 mm4	beamdimensions.com, CISC W150x22

J1 W150x22: elastic section modulus	Sx	159	x10^3 mm3	beamdimensions.com, CISC W150x22
J1 W150x22: mass	m	22.5	kg/m	beamdimensions.com, CISC W150x22
J1 self-weight (formula)	w_sw	0.220725	kN/m	Formula: mass x g / 1000
C1 HSS127x127x6.4: area	A	2960	mm2	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: moment of inertia	I	7.05	x10^6 mm4	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: radius of gyration	r	48.8	mm	CISC HSS table (search HSS127x127x6.4 on the same site)
C1 HSS127x127x6.4: mass	m	23.2	kg/m	CISC HSS table (search HSS127x127x6.4 on the same site)
BR1 L64x64x6.4: area	A	766	mm2	CISC angle table (bracing: model + MTO only, not designed)
BR1 L64x64x6.4: mass	m	6.01	kg/m	CISC angle table (bracing: model + MTO only, not designed)

LOADS	**Symbol**	**Value**	**Unit**	**Source / assumption**
Grating (dead)	q_D	0.5	kPa	Assumed
Floor live load	q_L	4.8	kPa	Assumed; replace with a cited value if found
Equipment point load (live), each main beam	P_eq	10	kN	Assumed
Guard load	-	cited only	-	Cite NBCC 2020 Div. B Part 4 guard clause / O. Reg. 213/91 guardrail section. NOT designed; flagged for review. Required per review R4; to be added at R1.
Interior joist tributary width	b_int	1.5	m	Formula: joist spacing
End joist tributary width	b_end	0.75	m	Formula: half joist spacing

LOAD COMBINATIONS / LIMITS	**Symbol**	**Value**	**Unit**	**Source / assumption**
Dead load factor (strength)	aD	1.25	-	1.25D + 1.5L
Live load factor (strength)	aL	1.5	-	1.25D + 1.5L
Deflection limit denominator (service, D+L)	L/	360	-	L/360 assumed

SCOPE LIMITS

1. Elastic Strength of Materials analysis only.
2. No CSA S16 resistance factors, no lateral-torsional buckling check, no column curve.
3. Connections, bracing and guard lateral load are not designed; flagged for senior review.
4. Loads are assumed unless a source is cited in column E.
5. Student exercise. Not for construction.

CHECK Three-way comparison

Quantity	Unit	Mathcad (hand)	Excel	Robot	% diff Excel vs hand	% diff Robot vs hand	PASS ($\leq$2%)	Cause if $>$2%
J1 max factored moment Mf	kN·m	13.51	13.515	13.51	0.0%	0.0%	PASS	Mathcad g = 9.807 m/s^2, Excel g = 9.81 m/s^2; under 0.05% effect on self-weight terms.
J1 max factored shear Vf	kN	18.02	18.020	18.02	0.0%	0.0%	PASS	
J1 service deflection	mm	3.561	3.561	3.56	0.0%	0.0%	PASS	
B1 reaction (factored)	kN	35.96	35.958	35.95	0.0%	0.0%	PASS	
B1 max factored moment Mf	kN·m	78.7	78.701	78.69	0.0%	0.0%	PASS	
B1 max factored shear Vf	kN	35.96	35.958	35.95	0.0%	0.0%	PASS	
B1 service deflection	mm	10.72	10.717	10.72	0.0%	0.0%	PASS	
C1 axial load (factored)	kN	46.03	46.028	46.83	0.0%	1.7%	PASS	Robot includes bracing self-weight (2.6 kN total; DL1 total 22.17 kN vs. 19.6 kN hand); +0.8 kN per post at ULS.
C1 Euler buckling load Pcr	kN	1546	1546.238	not run in robot	0.0%	N/A	N/A	
Total dead load	kN	19.6		22.17				bracing
Total live load	kN	106.4		106.4				

Review flags

- Long-side L64 braces sag about 24 mm under self-weight (L/280). Flag for senior review: consider a larger angle or a mid-length tie
- C1 member axial 41.9 kN (Robot) vs. 46.0 kN (hand): X-bracing shares about 4.9 kN of gravity load; the hand calc is conservative.

CALC Excel calculation (named inputs only)

J1 JOIST (interior)

Quantity	Value	Unit
wD dead UDL	0.971	kN/m
wL live UDL	7.200	kN/m
wf factored UDL	12.013	kN/m
Mf	13.515	kN·m
Vf	18.020	kN
sigma	85.001	MPa
tau	20.440	MPa
deflection (service)	3.561	mm
limit L/360	8.333	mm

B1 MAIN BEAM

Quantity	Value	Unit
PD per joist end	1.456	kN
PL per joist end	10.800	kN
Pf factored joist load	18.020	kN
Peq factored	15.000	kN
self-weight factored	0.476	kN/m
RB reaction	35.958	kN
MB midspan moment	78.701	kN·m
sigma	143.354	MPa
tau	19.999	MPa
Ps service joist load	12.256	kN
defl: joists at 1.5 and 4.5 m	4.456	mm
defl: midspan joist + equipment	5.884	mm
defl: self-weight	0.377	mm
deflection total	10.717	mm
limit L/360	16.667	mm

C1 POST (corner)

Quantity	Value	Unit
wD end joist	0.596	kN/m
wL end joist	3.600	kN/m
Re end joist reaction	9.217	kN
PC axial (factored)	46.028	kN
Pcr Euler	1546.238	kN
sigma axial	15.550	MPa
PC / Pcr	0.030	-

Values shown to 3 decimals; the live workbook (AS-P1-CAL-002.xlsx) holds full precision.

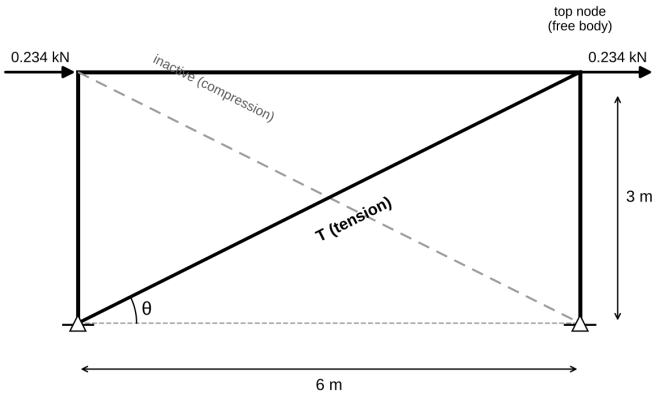
Purpose. Hand check of the R1 Robot model (AS-P1-MOD-001_R1) with BR1 braces (bars 18–25) set tension-only, under S16 notional loads and OBC guard loads, in response to review comments R1, R2 and R4. Robot reports axial force with compression positive; values below are tension unless noted.

Geometry and inputs. Long wall 6.0 m × 3.0 m: brace $L = \sqrt{6^2 + 3^2} = 6.708$ m, $\cos\theta = 0.894$, $\sin\theta = 0.447$, $\tan\theta = 0.5$. Short wall 3.0 m × 3.0 m: $L = 4.243$ m, $\cos\theta = \sin\theta = 0.707$, $\tan\theta = 1.0$. BR1 L64×64×6.4: A = 766 mm^2, 6.01 kg/m. C1: A = 2960 mm^2. E = 200 000 MPa. Gravity-only R1 post force (tension-only, ULS) = 46.55 kN.

1. S16 notional lateral load (Clause 8.4.1)

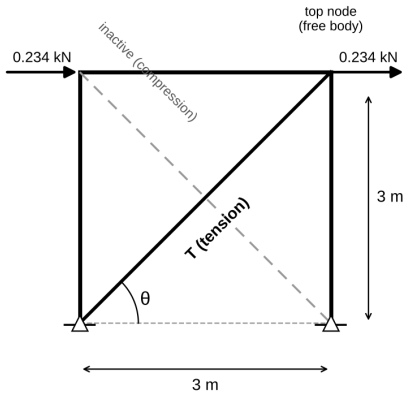
$N = 0.005 \times 187.32 = 0.94$ kN total (ULS gravity from Robot reactions), applied as 0.234 kN at each post top (nodes 5, 9, 10, 14). Load cases NX and NY, combined as 1.25D + 1.5L + 1.0N.

Long wall (6 m × 3 m), load in X



Brace length = √(6² + 3²) = 6.708 m
cos θ = 6 / 6.708 = 0.894
ΣFx at top node: T cos θ = 0.234 + 0.234 = 0.47 kN
T = 0.47 / 0.894 = 0.52 kN (Robot: 0.52)

Short wall (3 m × 3 m), load in Y



Brace length = √(3² + 3²) = 4.243 m
cos θ = 3 / 4.243 = 0.707
T cos θ = 0.47 kN
T = 0.47 / 0.707 = 0.66 kN (Robot: 0.66)

Each wall carries half the total (two parallel walls, symmetric frame): $H = 0.47$ kN per wall. At the loaded top node, $\Sigma F_x = 0$: $T\cos\theta = H$. Sway lengthens one diagonal of each X (tension, active); the other would shorten and is inactive.

	Long wall (NX)	Short wall (NY)
Brace tension $T = H/\cos\theta$	0.47/0.894 = **0.52 kN**	0.47/0.707 = **0.66 kN**
Brace self-weight $W = L \times 6.01 \times 9.81/1000$	0.395 kN	0.250 kN
Self-weight component along brace, ULS: $1.25\,W\sin\theta$	0.22 kN	0.22 kN
Tension at top end	0.52 + 0.22 = **0.74 kN**	0.66 + 0.22 = **0.88 kN**
Overturning $M = 0.94 \times 3.0 = 2.82$ kN·m; per post $M/b/2$	2.82/6/2 = 0.24 kN	2.82/3/2 = 0.47 kN
Post force $46.55 + \Delta$	**46.79 kN**	**47.02 kN**

2. Guard lateral load (OBC 4.1.5.14)

Guard load 0.75 kN/m outward at guard height (OBC 2024 Div. B, 4.1.5.14, “other locations”; category to be confirmed by reviewer). Applied as a horizontal line load on one edge only, which is the worst case for the bracing: GX on the 6 m-end joist (3 m long, bar 17); GY on one B1 line (6 m, bars 9–12). Combinations ULS+GX = 1.25D + 1.5L + 1.5GX + 1.0NX and ULS+GY = 1.25D + 1.5L + 1.5GY + 1.0NY. Combining full floor live load with guard load is conservative and is noted for review. The loaded edge member delivers its load to two posts, one on each parallel wall, so each wall carries half.

	ULS+GX (long walls)	**ULS+GY (short walls)**
Factored guard load $1.5 \times 0.75\times$ edge length	$1.125 \times 3 = 3.375$ kN	$1.125 \times 6 = 6.75$ kN
Total lateral (guard + notional 0.94)	**4.31 kN**	**7.69 kN**
Per wall H	2.16 kN	3.85 kN
Brace tension $T = H/\cos\theta$	2.16/0.894 = **2.41 kN**	3.85/0.707 = **5.44 kN**
Tension at top end (+ 0.22)	**2.63 kN**	**5.66 kN**
Overturning $M = H_{tot} \times 3.0$	12.94 kN·m	23.07 kN·m
Per post $M/b/2$	12.94/6/2 = 1.08 kN	23.07/3/2 = 3.85 kN
Post force $46.55 + \Delta$	**47.63 kN**	**50.40 kN**

3. SLS displacements (comment R1)

SLS combinations: SLS+GX = D + L + GX and SLS+GY = D + L + GY (unfactored, notional loads omitted).

- **Post shortening (gravity):** $\delta_p = PL/AE = 32.14 \times 10^3 \times 3000/(2960 \times 200\,000) = 0.16$ mm, using the SLS reaction 32.14 kN.
- **Brace slack.** Gravity lowers the post tops by δ_p, which shortens each diagonal by $\delta_p \sin\theta$. The tension brace only engages after the frame has swayed far enough to recover this: horizontal slack $= \delta_p \tan\theta$.
- **Brace stretch.** $\delta_b = TL/AE$; horizontal component $\delta_b/\cos\theta$.

	SLS+GX (long wall)	**SLS+GY (short wall)**
Service guard load per wall	$0.75 \times 3/2 = 1.125$ kN	$0.75 \times 6/2 = 2.25$ kN
Brace tension T	1.26 kN	3.18 kN
Brace stretch $\delta_b = TL/AE$	0.055 mm	0.088 mm
Horizontal: $\delta_b/\cos\theta$	0.062 mm	0.125 mm
Slack: $\delta_p \tan\theta$	0.16 × 0.5 = 0.081 mm	0.16 × 1.0 = 0.163 mm
Frame drift at post top	**0.14 mm**	**0.29 mm**

B1 lateral deflection (SLS+GY). The joists tie both B1 lines together, so they bend about their weak axes as a pair: $\delta = 5wL^4/(384E \cdot 2I_y) = 5 \times 0.75 \times 6000^4/(384 \times 200\,000 \times 2 \times 7.23 \times 10^6) = 4.38$ mm, plus 0.29 mm frame drift at the supports = **4.67 mm**. (I_y = 7.23×10^6 mm^4 for W310×39, CISC.)

B1 vertical deflection (SLS+GY). 10.72 mm (AS-P1-CAL-001) + 0.16 mm post shortening = **10.88 mm**.

4. Hand vs Robot

#	Quantity	Case	Hand	Robot	Diff
1	Brace T, long wall (bars 18, 20), bottom / top (kN)	ULS+NX	0.52 / 0.74	0.52 / 0.74	0.0%
2	Brace T, short wall (bars 22, 24), bottom / top (kN)	ULS+NY	0.66 / 0.88	0.66 / 0.88	0.0%
3	Post force (kN)	ULS+NX	46.79	46.79	0.0%
4	Post force (kN)	ULS+NY	47.02	47.02	0.0%
5	Brace T, long wall, bottom / top (kN)	ULS+GX	2.41 / 2.63	2.41 / 2.63	0.0%
6	Brace T, short wall, bottom / top (kN)	ULS+GY	5.44 / 5.66	5.43 / 5.66	0.2%
7	Post force (kN)	ULS+GX	47.63	47.63	0.0%
8	Post force (kN)	ULS+GY	50.40	50.40	0.0%
9	Post shortening (mm)	SLS	0.16	0.16	0.0%
10	Frame drift at post top (mm)	SLS+GX	0.14	0.15	−7%
11	Frame drift at post top (mm)	SLS+GY	0.29	0.30	−3%
12	B1 lateral at midspan (mm)	SLS+GY	4.67	4.65	+0.4%
13	B1 vertical at midspan (mm)	SLS+GY	10.88	10.88	0.0%

Drift differences of a few hundredths of a millimetre come from rounding and from minor flexibility not included by hand (beam axial shortening, connection eccentricities); they are not significant.

5. Results carried forward

- Under gravity alone all braces are inactive and the posts carry the full load (46.55 kN). The R0 finding of about 4.9 kN through the bracing was a result of the elastic model and is closed (R2).
- Under lateral load exactly one brace per X is active, in tension, and every force above is confirmed by hand (R1, R4).
- **Governing brace force for AS-P1-CAL-004: T_f = 5.66 kN** (bars 22/24, ULS+GY).
- **Governing post force for AS-P1-CAL-004: C_f = 50.40 kN** (ULS+GY).
- SLS frame drift 0.30 mm (H/10 000); B1 lateral 4.65 mm (L/1290). Both are far below any applicable limit.

AS-P1-CAL-004 | Rev R1 | A. Saleh | 2026-10-21
CSA S16-19 member checks: B1, J1, C1, BR1.
Student exercise, not for construction.

$F_y := 350 \cdot MPa$
$E := 200000 \cdot MPa$ 350W assumed (CSA G40.21). S16 Cl. 13.1
$\phi := 0.9$
$F_u := 450 \cdot MPa$
$G := 77000 \cdot MPa$
$\phi_u := 0.75$

B1, W310×39 (Cl. 11, 13.4.1.1, 13.6, 13.8)

$d_B1 := 310 \cdot mm$ $b_B1 := 165 \cdot mm$ $t_B1 := 9.7 \cdot mm$
$w_B1 := 5.8 \cdot mm$
$Z_x.B1 := 610 \cdot 10^3 \cdot mm^3$
$I_y.B1 := 7.27 \cdot 10^6 \cdot mm^4$
$J_B1 := 126 \cdot 10^3 \cdot mm^4$
$C_w.B1 := 164 \cdot 10^9 \cdot mm^6$

$A_B1 := 4940 \cdot mm^2$
$I_x.B1 := 85.1 \cdot 10^6 \cdot mm^4$

$$Fy_n := \frac{F_y}{MPa}$$

$$flangeB1 := \frac{b_B1}{(2 \cdot t_B1)}$$

$flangeB1 = 8.505$

$$limit1 := \frac{145}{\sqrt[2]{Fy_n}}$$

$limit1 = 7.751$

$$limit2 := \frac{170}{\sqrt[2]{Fy_n}}$$

$limit2 = 9.087$

$hB1 := d_B1 - 2 \cdot t_B1$

$$webB1 := \frac{hB1}{w_B1}$$

$webB1 = 50.103$

$$limitw1 := \frac{1100}{\sqrt[2]{Fy_n}}$$

$$limitw1 = 58.797$$

Flange Class 2, web Class 1, so section Class 2 (governed by the worse of the two). Use Z_x for Mr

$$limitV := \frac{1014}{\sqrt[2]{Fy_n}} \qquad limitV = 54.201$$

$$VrB1 := \phi \cdot d_B1 \cdot w_B1 \cdot 0.66 \cdot F_y \qquad VrB1 = 373.804\ \mathbf{kN}$$

$$MfB1 := 78.70 \cdot \mathbf{kN} \cdot \mathbf{m}$$
$$VfB1 := 35.96 \cdot \mathbf{kN}$$

$$ratioVB1 := \frac{VfB1}{VrB1} \qquad ratioVB1 = 0.096$$

$$MpB1 := Z_x.B1 \cdot F_y \qquad MpB1 = 213.5\ \mathbf{kN} \cdot \mathbf{m}$$

$$omega2 := 1 \qquad omega2 = 1$$

$$LuB1 := 1.5 \cdot \mathbf{m} \qquad LuB1 = 1.5\ \mathbf{m}$$

$$term1 := E \cdot I_y.B1 \cdot G \cdot J_B1$$

$$k := \pi \cdot \frac{E}{LuB1}$$

$$term2 := k \cdot k \cdot I_y.B1 \cdot C_w.B1$$

$$MuB1 := omega2 \cdot \left(\frac{\pi}{LuB1}\right) \cdot \sqrt[2]{term1 + term2}$$

$$MuB1 = 989.707\ \mathbf{kN} \cdot \mathbf{m}$$

$$MrB1 := \text{if}\left(MuB1 > 0.67 \cdot MpB1,\ min\left(1.15 \cdot \phi \cdot MpB1 \cdot \left(1 - 0.28 \cdot \frac{MpB1}{MuB1}\right),\ \phi \cdot MpB1\right),\ \phi \cdot MuB1\right)$$

$MrB1 = 192.15\ \ kN \cdot m$

$$ratioMB1 := \frac{MfB1}{MrB1}$$

$ratioMB1 = 0.41$

$LuU := 6 \cdot m$

$$kU := \pi \cdot \frac{E}{LuU}$$

$term2U := kU \cdot kU \cdot I_y.B1 \cdot C_w.B1$

$$MuB1u := omega2 \cdot \left(\frac{\pi}{LuU}\right) \cdot \sqrt[2]{term1 + term2U}$$

$MuB1u = 86.325\ \ kN \cdot m$

$$MrB1u := \text{if}\left(MuB1u > 0.67 \cdot MpB1,\ min\left(1.15 \cdot \phi \cdot MpB1 \cdot \left(1 - 0.28 \cdot \frac{MpB1}{MuB1u}\right),\ \phi \cdot MpB1\right),\ \phi \cdot MuB1u\right)$$

$MrB1u = 77.692\ \ kN \cdot m$

$$ratioMB1u := \frac{MfB1}{MrB1u}$$

$ratioMB1u = 1.013$

If J1 does not restrain the B1 compression flange, Lu = 6.0 m and B1 fails (1.01). Lateral restraint at the joists is required (Finding 8).

$Z_y.B1 := 135 \cdot 10^3 \cdot mm^3$

$MryB1 := \phi \cdot Z_y.B1 \cdot F_y \qquad MryB1 = 42.525\ \ kN \cdot m$

$$wG := 1.125 \cdot \frac{kN}{m}$$

$L6 := 6 \cdot m$

$MfyB1 := wG \cdot L6 \cdot \frac{L6}{8}$ $MfyB1 = 5.063\ kN \cdot m$

$rx := \frac{MfB1}{MrB1}$

$ry := \frac{MfyB1}{MryB1}$

$ratioBiax := rx + ry$

$ratioBiax = 0.529$

J1, W150×22 (Cl. 11, 13.5(c), 13.4.1.1, 13.6

$d_J1 := 152 \cdot mm$ $b_J1 := 152 \cdot mm$ $t_J1 := 6.6 \cdot mm$ $w_J1 := 5.8 \cdot mm$

$I_x.J1 := 12.1 \cdot 10^6 \cdot mm^4$ $I_y.J1 := 3.87 \cdot 10^6 \cdot mm^4$

$J_J1 := 43.4 \cdot 10^3 \cdot mm^4$ $C_w.J1 := 20.4 \cdot 10^9 \cdot mm^6$

$MfJ1 := 13.51 \cdot kN \cdot m$ $VfJ1 := 18.02 \cdot kN$

$flangeJ1 := \frac{b_J1}{(2 \cdot t_J1)}$ $flangeJ1 = 11.515$

$limit3 := \frac{200}{\sqrt[2]{Fy_n}}$ $limit3 = 10.69$

$hJ1 := d_J1 - 2 \cdot t_J1$

$webJ1 := \frac{hJ1}{w_J1}$ $webJ1 = 23.931$

Flange Class 4 → effective flange width per S16 Cl. 13.5(c).

$be := 400 \cdot \frac{t_J1}{\sqrt[2]{Fy_n}}$ $be = 141.114\ mm$

$dA := (b_J1 - be) \cdot t_J1$

$$yf := \frac{(d_J1 - t_J1)}{2}$$

$$Ie := I_xJ1 - 2 \cdot dA \cdot yf \cdot yf$$

$$Se := \frac{Ie}{\left(\frac{d_J1}{2}\right)} \qquad Se = \left(1.492 \cdot 10^{5}\right)\ mm^{3}$$

$$MyJ1 := Se \cdot F_y \qquad MyJ1 = 52.226\ kN \cdot m$$

$$VrJ1 := \phi \cdot d_J1 \cdot w_J1 \cdot 0.66 \cdot F_y \qquad VrJ1 = 183.285\ kN$$

$$ratioVJ1 := \frac{VfJ1}{VrJ1} \qquad ratioVJ1 = 0.098$$

$$LuJ1 := 3 \cdot m$$

$$term1J := E \cdot I_yJ1 \cdot G \cdot J_J1$$

$$kJ := \pi \cdot \frac{E}{LuJ1}$$

$$term2J := kJ \cdot kJ \cdot I_yJ1 \cdot C_wJ1$$

$$MuJ1 := omega2 \cdot \left(\frac{\pi}{(LuJ1)}\right) \cdot \sqrt[2]{term1J + term2J} \qquad MuJ1 = 81.45\ kN \cdot m$$

$$MrJ1 := \text{if}\left(MuJ1 > 0.67 \cdot MyJ1,\ min\left(1.15 \cdot \phi \cdot MyJ1 \cdot \left(1 - 0.28 \cdot \frac{MyJ1}{MuJ1}\right),\ \phi \cdot MyJ1\right),\ \phi \cdot MuJ1\right)$$

$$MrJ1 = 44.349\ kN \cdot m$$

$$ratioMJ1 := \frac{MfJ1}{MrJ1} \qquad ratioMJ1 = 0.305$$

Class 4: My = Se·Fy used in place of Mp. J1 taken as unbraced over 3.0 m (no restraint from grating).

C1, HSS127×127×6.4 (Cl.13.3.1):

$A_C1 := 2960 \cdot mm^2$ $r_C1 := 48.8 \cdot mm$ $b_C1 := 127 \cdot mm$ $t_C1 := 6.35 \cdot mm$

$CfC1 := 50.40 \cdot kN$ $L_C1 := 3 \cdot m$ $K := 1$

$belt := \frac{(b_C1 - 4 \cdot t_C1)}{t_C1}$ $belt = 16$

$KLr := K \cdot \frac{L_C1}{r_C1}$ $KLr = 61.475$

$ratioFe := \frac{Fy_n}{(\pi \cdot \pi \cdot 200000)}$

$Slend := KLr \cdot \sqrt[2]{ratioFe}$

$Slend = 0.819$

$n := 1.34$

$p1 := \exp(n \cdot 2 \cdot \ln(Slend))$

$base := 1 + p1$

$fac := \exp\left(\frac{-\ln(base)}{n}\right)$

$CrC1 := \phi \cdot A_C1 \cdot F_y \cdot fac$

$CrC1 = 661.244\ kN$

$ratioC1 := \frac{CfC1}{CrC1}$

$ratioC1 = 0.076$

Cr = φAFy(1 + λ^2n)^(-1/n), n = 1.34 (Class C HSS), S16 Cl. 13.3.1. Written with exp/ln in Mathcad.

BR1, L64×64×6.4 and M20 A325 bolt (Cl. 13.2, 12.3.3, 13.12.1.2)

$A_BR := 768 \cdot mm^2$

$t_BR := 6.35 \cdot mm$

$TfBR := 5.66 \cdot kN$

$F_ub := 825 \cdot MPa$

$A_b := 314 \cdot mm^2$

$d_b := 20 \cdot mm$

$TrG := \phi \cdot A_BR \cdot F_y$

$TrG = 241.92\ kN$

$An := A_BR - 24 \cdot mm \cdot t_BR$
$Ane := 0.60 \cdot An$

$TrN := \phi_u \cdot Ane \cdot F_u$

$TrN = 124.659\ kN$

$VrBolt := 0.6 \cdot 0.8 \cdot A_b \cdot F_ub \cdot 0.7$

$VrBolt = 87.041\ kN$

$BrBolt := 3 \cdot 0.8 \cdot t_BR \cdot d_b \cdot F_u$

$BrBolt = 137.16\ kN$

$ratioBR := \frac{TfBR}{VrBolt}$

$ratioBR = 0.065$

Assumed 1 M20 A325 bolt per brace end, threads intercepted. Shear lag 0.60An (angle connected by one leg, fewer than 4 bolts). Bolt shear governs.

SUMMARY: CSA S16 MEMBER CHECKS	Factored	Resistance	Ratio	
B1 Bending, Lu = 1.5 m	78.70 kN·m	192.2 kN·m	0.41	
B1 Shear	35.96 kN	373.8 kN	0.10	
B1 Biaxial bending (ULS+GY)			0.53	
B1 Bending, Lu = 6.0 m (unbraced)	78.70 kN·m	77.7 kN·m	1.01	FAILS: needs J1 restraint
J1 Bending, Class 4, Lu = 3.0 m	13.51 kN·m	44.3 kN·m	0.30	
J1 Shear	18.02 kN	183.3 kN	0.10	
C1 Compression	50.40 kN	661.2 kN	0.08	
BR1 Tension, gross	5.66 kN	241.9 kN	0.02	
BR1 Tension, net	5.66 kN	124.7 kN	0.05	
BR1 Bolt shear (governs)	5.66 kN	87.1 kN	0.07	
BR1 Bolt bearing	5.66 kN	137.2 kN	0.04	

All members pass with J1 providing lateral restraint to B1.

AS-P1-MTO-001 Material Take-Off, Maintenance Platform (Portfolio Study)

Rev R1, 2026-10-21. Lengths from Revit schedules (AS-P1-MOD-002, sheet S-301). Mass/m from CISC Handbook. Grey cells = inputs.

Mark	Member	Section (CISC)	Revit type	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
B1	Main beam	W310x39	W12X26	2	5.848	11.696	38.80	453.8	Revit cut length
J1 (end)	Joist	W150x22	W6X15	2	2.848	5.696	22.50	128.2	J1-1, J1-5
J1 (int.)	Joist	W150x22	W6X15	3	2.810	8.430	22.50	189.7	J1-2 to J1-4
C1	Post	HSS127x127x6.4	HSS5X5X1/4	4	3.000	12.000	23.20	278.4	System length GRADE to T/S
BR1 (long)	X-brace, long sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	6.463	25.852	6.01	155.4	BR1-1 to BR1-4
BR1 (short)	X-brace, short sides	L64x64x6.4	L2-1/2X2-1/2X1/4	4	3.848	15.392	6.01	92.5	BR1-5 to BR1-8
Subtotal, members				**19**		**79.066**		**1297.9**	
Connection allowance	Plates, bolts, welds						10%	129.8	ASSUMED % of member mass, not from code. Confirm with reviewer.
TOTAL STEEL								**1427.7**	**kg**

Cross-check against Revit schedule volumes

Item	Value	Unit	Source
Framing volume	0.1336	m3	P1 Steel Framing Schedule, grand total
Column volume	0.0355	m3	P1 Column Schedule, grand total
Steel density	7850.0	kg/m3	Standard value for structural steel
Mass from Revit volume	1327.4	kg	Formula
Mass from CISC kg/m	1297.9	kg	Members subtotal above
Difference	2.3%		Expected small: Revit families use imperial section geometry
Status	WITHIN 5% - OK		Tolerance 5% (own criterion)

BR1 Slenderness Check, L64×64×6.4 X-Bracing

Lengths are work-point (centre-to-centre) geometry, conservative versus Revit cut length. Grey cells = inputs.

r_min (mm)	12.5	Computed: $\sqrt{}$(Iyp/A), Iyp and A from beamdimensions.com L64x64x6.4
Tension limit L/r	300	CSA S16 slenderness limit, tension members
Compression limit KL/r	200	CSA S16 slenderness limit, compression members

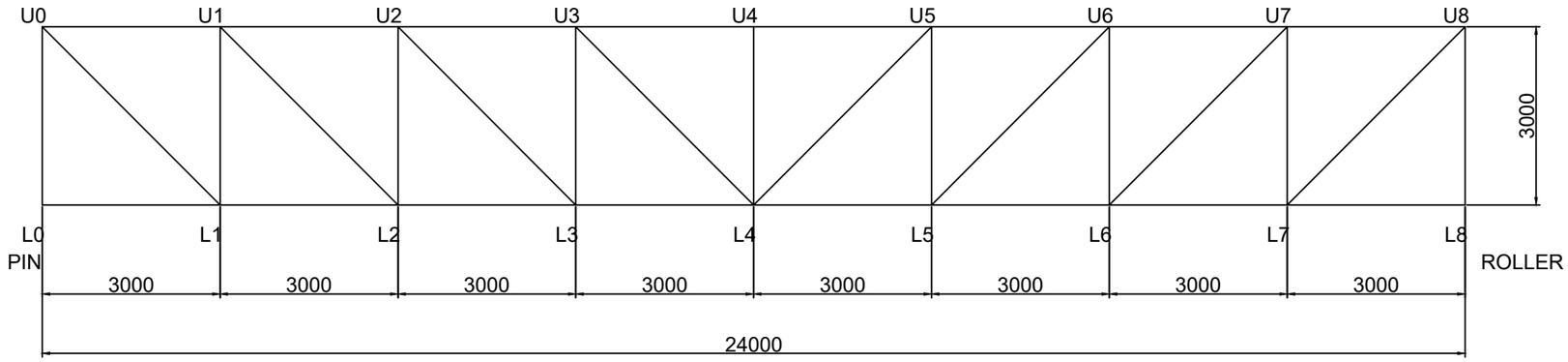
Brace	Panel w (mm)	Panel h (mm)	L full (mm)	L/r full	L at crossing (mm)	L/r at crossing	Tension-only check	If designed for compression
Long sides (BR1-1 to 4)	6000	3000	6708	537	3354	268	OK if bolted at crossing	FAIL - keep tension-only
Short sides (BR1-5 to 8)	3000	3000	4243	339	2121	170	OK if bolted at crossing	OK

Finding: Full-length L/r exceeds the tension limit on both brace types. Braces must be bolted at the X intersection (halves unbraced length) and treated as tension-only. Add note to S-101 and brief. (Done at R1: tension-only bracing, bolted crossing, half-length L/r 268 and 170, both $\leq$ 300.)

AS-P1-REG-001 Review and Issue Register

Rev R1, 2026-10-21

No.	Date	Source	Item	Finding	Action	Status
1	2026-10-18	Revit schedule	BR1 section	Braces modelled as W6X15. Schedule volume check gave ∼2950 mm2 area vs ∼768 mm2 expected for L64x64x6.4.	Type changed to L2-1/2X2-1/2X1/4 (L64x64x6.4 equivalent). Schedule re-checked.	Closed
2	2026-10-18	Revit schedule	J1 structural usage	Joists tagged as Girder.	Changed to Joist.	Closed
3	2026-10-18	Revit model	BR1 placement method	Braces drawn with Beam tool, so Structural Usage cannot be set to Vertical Bracing.	Accepted for this model. Use Brace tool on future models.	Closed
4	2026-10-18	Drawing S-101 R0	C1 label	Section labelled HSS128x128.	Corrected to HSS127x127x6.4 in R1.	Closed
5	2026-10-18	Drawing S-101 R0	BR1 label	Typo 'BR!'.	Corrected to BR1 in R1.	Closed
6	2026-10-18	Drawing S-101 R0	T/S label	Top-of-steel label poorly placed.	Moved in R1.	Closed
7	2026-10-18	Robot vs hand calc	C1 reaction	Robot 46.83 kN vs hand 46.03 kN.	Difference = bracing self-weight, included in Robot only. Accepted.	Closed
8	2026-10-18	Robot vs hand calc	C1 axial force	Robot member force 41.9 kN, lower than hand calc.	R1 tension-only run: braces inactive under gravity; post force 46.55 kN. Gap was elastic brace participation.	Closed
9	2026-10-18	Slenderness check	BR1 L/r	Full-length L/r exceeds tension limit (see SLENDERNESS tab).	Tension-only adopted; half-length L/r 268 $\leq$ 300. Noted on S-101/S-201 Rev 2–3.	Closed
10	2026-10-18	Robot	BR1 self-weight sag	Brace sag ∼24 mm (about L/280) under self-weight.	Intersection connection reduces it. Mention in brief.	Open
11	2026-10-18	MTO	Mass cross-check	Revit volume mass vs CISC kg/m mass differ by a few %.	Imperial family geometry. Within 5% tolerance. Accepted.	Closed
12	2026-10-18	Revit type	C1 section area	Earlier note said Revit HSS area 2774 mm2 vs CISC 2960. Column volume implies ∼2967 mm2.	Checked: column schedule volume 0.0089 m3 / 3.0 m = 2967 mm2, matches CISC 2960 mm2.	Closed
13	2026-10-21	CISC check	BR1 area	R0 used 766 mm2; CISC gives 768 mm2 (2L64x64x6.4 = 1536 mm2).	R1 uses 768 mm2. Difference 0.3%, no effect on R0 results.	Closed
14	2026-10-21	S16 check (CAL-004)	J1 flange class	b/2t = 11.5 exceeds Class 3 limit 200/sqrt(Fy) = 10.7; J1 flange is Class 4 in 350W.	Effective flange width 400t/sqrt(Fy) = 141 mm used (S16 Cl. 13.5(c)); Mr = 44.3 kN.m, utilization 0.30.	Closed
15	2026-10-21	S16 check (CAL-004)	B1 lateral support	B1 Mr assumes compression flange braced by J1 at 1.5 m (utilization 0.41). Unbraced over 6.0 m, utilization 1.01.	Joist-to-beam connection to provide lateral restraint; confirm in connection design.	Open



CONVEYOR GALLERY TRUSS - ELEVATION 1:100

1. TRIAL SECTIONS: CHORDS HSS102x102x6.4; VERTICALS AND DIAGONALS L76x76x6.4. NOT DESIGN-CHECKED.
2. LOADS (ASSUMED): DEAD 2.0 kN/m, LIVE 2.4 kN/m. ULS 1.25D + 1.5L = 18.3 kN AT L1-L7.
3. MEMBER FORCES: SEE AS-P2-CAL-001 (HAND CHECK) AND AS-P2-CAL-002 (VERIFIED AGAINST ROBOT AS-P2-MOD-001).
4. CONNECTIONS NOT DESIGNED. STUDENT EXERCISE. NOT FOR CONSTRUCTION.

CONVEYOR GALLERY TRUSS VERIFICATION

S-301 TRUSS ELEVATION

SCALE 1:100 DRAWN A. SALEH 2026-10-18

STUDENT EXERCISE. NOT FOR CONSTRUCTION

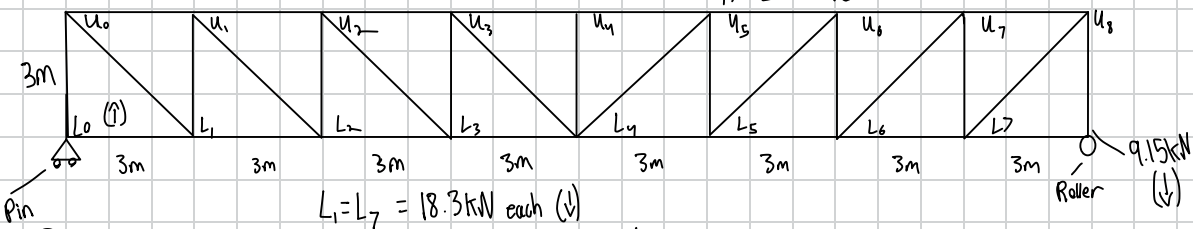
REV 0 2026-10-18 FIRST ISSUE

P2 Conveyor Gallery Truss

Sign Convention: Tension (+), Compression (-)

73.2 kN (↑) at L0 and L8

No loads U0-U8. Horizontal reaction L0 = 0



Determinacy check

33 members + 3 reactions = 36 = 2x18 nodes.

Therefore the truss is determinate.

Assumed Loads

Dead: 1.5 kN/m conveyor + 0.5 kPa x 1m grating = 2 kN/m

Live: 2.4 kPa x 1.0 m = 2.4 kN/m

ULS: 1.25x2 + 1.5x2.4 = 6.1 kN/m

Nodal loads: 18.3kN down at L1-L7, and 9.15 kN at L0 and L8

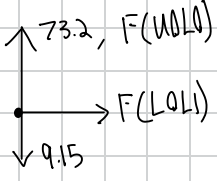
Diagonal Length 4.243m, Sin = Cos = 0.7071

Reactions

Total load = 6.1 x 24 = 146.4 kN

Each reaction = $\frac{146.4}{2}$ = 73.2

Joint L0



$\sum F_x = 0$: F(L0L1) = 0, so L0-L1 = 0 (zero force)

$\sum F_y = 0$: 73.2 - 9.15 + F(U0L0) = 0, so F(U0L0) = -64.05, which is 64.05 kN (C)

Joint U0



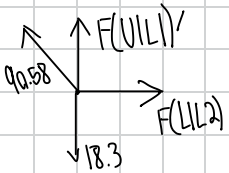
$\sum F$

$\sum F_y = 0$: 64.05 - 0.7071 · F(U0L1) = 0, So F(U0L1) = 90.58 kN (T)

$\sum F_x = 0$: F(U0U1) + 0.707 (90.58) = 0, So F(U0U1) = -64.05 kN

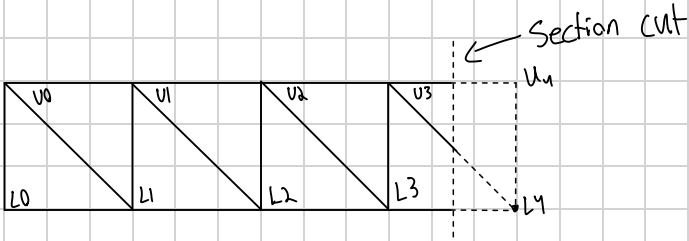
= 64.05 kN (C)

Joint L1



$\sum F_y = 0: 0.7071(90.58) + F(U1L1) - 18.3 = 0$, So $F(U1L1) = -45.75 \rightarrow 45.75\,kN(C)$

$\sum F_x = 0: -0.7071(90.58) + F(L1L2) = 0$, So $F(L1L2) = 64.05\,kN(T)$



$\sum M_{L4} = 0: 73.2(12) - 9.15(12) - 18.3(9+6+3) - F(U3U4)\cdot(-3) = 0$

$F(U3U4) = \frac{-439.2}{3} = -146.40 \rightarrow 146.40\,kN(C)$

Full member force table (hand results above verified against robot model)

Group	Member	Mirror (right half)	Length (m)	Force (kN)	T / C
Top chord	U0–U1	U7–U8	3.000	−64.05	C
	U1–U2	U6–U7	3.000	−109.80	C
	U2–U3	U5–U6	3.000	−137.25	C
	U3–U4	U4–U5	3.000	−146.40	C
Bottom chord	L0–L1	L7–L8	3.000	0.00	0 (zero-force)
	L1–L2	L6–L7	3.000	+64.05	T
	L2–L3	L5–L6	3.000	+109.80	T
	L3–L4	L4–L5	3.000	+137.25	T
Vertical	U0–L0	U8–L8	3.000	−64.05	C
	U1–L1	U7–L7	3.000	−45.75	C
	U2–L2	U6–L6	3.000	−27.45	C
	U3–L3	U5–L5	3.000	−9.15	C
	U4–L4	—	3.000	0.00	0 (zero-force)
Diagonal	U0–L1	U8–L7	4.243	+90.58	T
	U1–L2	U7–L6	4.243	+64.70	T
	U2–L3	U6–L5	4.243	+38.82	T
	U3–L4	U5–L4	4.243	+12.94	T

AS-P2-CAL-002 Hand vs Robot member forces, Conveyor Gallery Truss (ULS 1.25D + 1.5L, self-weight excluded)

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction. Robot FX sign: + = compression; reversed below to match hand convention (+ = tension).

Bar	Member	Hand (kN)	Robot FX (kN)	Robot, hand sign (kN)	Difference (kN)	% diff	Status	Hand source
1	L0–L1	0.00	0.00	0.00	0.00	0.0%	PASS	Joints L0/U0/L1
2	L1–L2	64.05	-64.05	64.05	0.00	0.0%	PASS	Joints L0/U0/L1
3	L2–L3	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
4	L3–L4	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
5	L4–L5	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
6	L5–L6	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
7	L6–L7	64.05	-64.05	64.05	0.00	0.0%	PASS	Robot (symmetry check)
8	L7–L8	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
9	U0–U1	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
10	U1–U2	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
11	U2–U3	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
12	U3–U4	-146.40	146.40	-146.40	0.00	0.0%	PASS	Section cut, panel 4
13	U4–U5	-146.40	146.40	-146.40	0.00	0.0%	PASS	Robot (symmetry check)
14	U5–U6	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
15	U6–U7	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
16	U7–U8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
17	U0–L0	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
18	U1–L1	-45.75	45.75	-45.75	0.00	0.0%	PASS	Joints L0/U0/L1
19	U2–L2	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
20	U3–L3	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
21	U4–L4	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
22	U5–L5	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
23	U6–L6	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
24	U7–L7	-45.75	45.75	-45.75	0.00	0.0%	PASS	Robot (symmetry check)
25	U8–L8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
26	U0–L1	90.58	-90.58	90.58	0.00	0.0%	PASS	Joints L0/U0/L1
27	U1–L2	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
28	U2–L3	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
29	U3–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
30	U5–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
31	U6–L5	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
32	U7–L6	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
33	U8–L7	90.58	-90.58	90.58	0.00	0.0%	PASS	Robot (symmetry check)

Reactions: hand 73.20 kN each; Robot 73.20 kN each (sum 146.40 kN). Max force 146.40 kN = M/d check.

Members with hand source 'Robot (symmetry check)' were not individually hand-solved; they satisfy symmetry and joint U4 equilibrium.

AS-P2-MTO-001 Material Take-Off: Conveyor Gallery Truss Verification

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Based on S-301 Rev 0; trial sections; centreline lengths. Student exercise, not for construction.

Mark	Member	Section (CISC)	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
TC	Top chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	U0-U1 to U7-U8
BC	Bottom chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	L0-L1 to L7-L8
V	Vertical	L76x76x6.4	9	3.000	27.00	7.28	196.6	U0-L0 to U8-L8
D	Diagonal	L76x76x6.4	8	4.243	33.94	7.28	247.1	U0-L1 to U3-L4, U5-L4 to U8-L7
Subtotal, one truss			**33**		**108.94**		**1317.3**	
Connection allowance	Gusset plates, bolts, welds					10%	131.7	10% as in AS-P1-MTO-001; replace once connections designed
TOTAL, ONE TRUSS							**1449.0**	**kg**
TOTAL, TWO TRUSSES			**2**				**2898.0**	**kg (assumed one truss each side of gallery)**

Checks

Item	Value	Target	Status
Member count	33	33	OK - matches 2j - r = 33
Chord length, each chord (m)	24.00	24	OK - equals span
Vertical count	9	9	OK - 8 panels + 1
Diagonal length (m)	4.243	4.243	OK - 3 m panel x 3 m depth

Mass per metre = A x 7850 kg/m3, with A from CISC section properties (HSS102x102x6.4: 2320 mm2; L76x76x6.4: 927 mm2), same method as AS-P1-MTO-001. Lengths are node-to-node centreline lengths from S-301; no allowance for cut lengths or end preparation.

Maintenance Access in Hydropower Plants: Permanent Platforms and Scaffolding

AS-P3-RPT-001

Revision 0 18 October 2026

Literature Review Prepared by: Abbas Saleh

Research Question

For turbine maintenance which has to be done over and over again, what are the practical differences between using a permanent modular platform and setting up temporary scaffolding each time? This review looks at the fall hazards which come with each system, the amount of time which is needed to get access set up before the work can start, and the inspection and maintenance work which comes with each system.

Summary

From the sources which were reviewed, the main benefit of a permanent platform does not seem to come from the inspection itself. It comes from the work which has to happen before and after each inspection. The scaffold has to be built, checked, and then taken down again, and this is repeated every time access is needed. At the same time, the workers who are doing this are exposed to fall hazards while the scaffold is still incomplete. The sources which were found suggest that a permanent arrangement would be safer and would also take less time to get onto. However, no comparison was found where the hours saved or the incidents avoided were actually measured, and there was also very little information on what is needed to maintain a permanent platform over its life. Because of this, the case for a permanent platform at the moment is based on the reduction in repeated scaffold work rather than on any number which can be quoted.

1. Background

In many cases, inspecting the underside of a turbine runner is more of an access problem than an inspection problem. The work has to be done inside the draft tube, where there is not much room to move around. The usual approach is to build scaffolding inside the draft tube elbow so that the inspection areas can then be reached. After the work is finished, the scaffold is then removed. If access is needed again during a later outage, the same arrangement then has to be built again from the beginning.

Hatch (2026) published a different approach in January 2026 [1]. A modular access platform was developed for SCHM's McCormick Hydropower Plant in Québec. The project started in 2019 after a safety concern was found underneath the turbine runner during an inspection. From the information which Hatch provided, the finished system serves five turbine-generator units, works with three different turbine types, and removes the need for conventional scaffolding.

The reason this project was chosen is because turbine maintenance is repeated. A temporary solution may be fine when access is only needed once in a while. It becomes a lot less attractive when essentially the same working area has to be set up again every time the equipment is inspected. At the same time, this does not automatically mean that the permanent installation is the better option. It still has to be designed for the site, inspected, and also maintained. Because of this, what needs to be looked at is whether the amount of scaffold work which is avoided is actually worth more than the extra responsibilities which then come with the permanent platform.

2. Sources and Scope

Six publicly available sources were reviewed. These include Ontario safety legislation, provincial working-at-heights guidance, a University of Toronto scaffold standard, one peer-reviewed study, and project information which was published by Hatch and by ZARGES. The legislation and safety documents were used to find the requirements which apply to scaffolding and fall protection. The Hatch and ZARGES publications were then

used to look at the alternatives to conventional access which are currently being used in hydropower stations.

There is a limitation with using company publications for this kind of comparison. The companies explain what their own systems are supposed to do. At the same time, the operating records which would be needed to actually check those benefits are not included, so the claims cannot be confirmed from the publications on their own. Ontario Regulation 213/91 was also used as the reference regulation. This is because it has specific requirements in it for scaffolding and also for fall protection. It is not being presented as the regulation which governs McCormick, since McCormick is located in Québec. This review also did not try to verify the structural capacity of either access system or estimate the cost of building a permanent platform. Both of these would have to be done separately.

Table 1. Information available from the reviewed sources

Reference	Relevant information	What remains unclear
Hatch [1]	Describes the McCormick platform and the problems which come with repeated scaffolding	Labour hours, incident data and maintenance requirements
Ontario guidance [2]	Explains mandatory working-at-heights training	Does not compare access systems
O. Reg. 213/91 [3]	Gives the requirements which apply to fall protection, guardrails and scaffolding	Does not say anything about replacing scaffolding with a permanent platform
UofT EHS [4]	Gives scaffold erection and inspection guidance	Does not address permanent turbine platforms
Robson et al. [5]	Reports changes in fall-related injury claims after the Ontario training requirement	Does not measure the effect of permanent platforms
ZARGES [6]	Describes quick setup of portable access equipment at a hydropower station	Not directly comparable to McCormick's permanent system

3. Discussion

3.1 Fall Hazards

When scaffolding is being considered, it is easy to only think about the finished platform which the workers end up standing on. However, someone then has to build it first. During erection, workers may be at height before the whole scaffold and its protective systems are complete. The same problem then comes up again during dismantling. These activities are separate from the turbine inspection, but they still expose workers to hazards just so that the inspection can take place.

There are a few parts of Ontario Regulation 213/91 which then apply to this kind of work [3]. In Section 26, fall protection is required where a worker could fall more than three metres. In Section 26.3, guardrails are required along scaffold and platform edges where the distance which the worker could fall is 2.4 metres or more. Section 26.1 then also states that guardrail systems should be used ahead of the personal fall protection methods. This difference matters here, since a worker who is standing on a finished and properly guarded platform is in a very different situation from a worker who is assembling access equipment while some of that protection is not in place yet. The University of Toronto scaffold standard also covers protection for workers who are doing erection work at height for the same reason [4]. In addition, the mandatory working-at-heights training in Ontario applies to workers who use fall protection devices such as travel restraint and fall arrest systems, and guardrails are not included in that list [2], which is consistent with the preference for guardrail systems which is given in Section 26.1 [3].

Taking this into consideration, a permanent platform with proper guarding could avoid the repeated exposure which happens during scaffold assembly and removal. This would be a meaningful advantage for a station

which needs access during several outages. It would not make the work free of fall hazards though. The platform would still need to provide safe access, proper edge protection, and also protection for any activity which takes a worker outside of the guarded area.

Robson et al. (2024) give some useful background on fall prevention [5]. Their study examined the mandatory working-at-heights training in Ontario. From their results, the lost-time claim rate for targeted fall injuries was 19% lower in 2017–19 than in 2012–14, compared to a 6% reduction in other provinces over the same period. These percentages should not be taken as the expected improvement from installing a permanent platform, since the study looked at training and not at alternative access structures. The more defensible conclusion is a narrower one. Avoiding scaffold erection and dismantling removes two recurring activities where fall exposure can happen. The actual reduction in risk would then have to be worked out for the specific station and work procedure.

3.2 Time Required to Establish Access

Time becomes a more important consideration when the same access equipment has to be installed again and again. For a scaffold arrangement, the work does not start when the inspection crew steps onto the platform. The components first have to be brought into position, assembled, and then inspected. Once the maintenance is finished, the structure then has to be dismantled and removed as well. At a generating station this work also has to fit within the planned outage. For this reason, the time which is spent setting up access is worth looking at separately from the inspection itself.

Hatch (2026) identifies scaffold installation as a time-consuming part of the conventional method [1]. However, the article does not give the crew size, the number of hours which are normally needed, or the effect on the outage schedule. Because of this, the size of the saving at McCormick cannot be worked out from it.

ZARGES gives another example from the Töging am Inn hydropower station in Germany [6]. The station has fourteen turbines, and one unit goes through a major overhaul each year over approximately three months. Portable work platforms were used in place of scaffolding for components up to four metres high, and ZARGES states that the equipment can be ready to use in less than a minute. This number needs some context. The ZARGES equipment is a portable access system. It is not the same as the permanent modular arrangement which was developed for McCormick, and so the setup time cannot be carried over from one system to the other. Both projects show why alternatives to scaffolding are being considered, but neither one gives enough data to calculate the time which would actually be saved.

A useful comparison would start from the existing maintenance procedure. Some examples of what would be needed are the number of workers who are assigned to scaffold erection, how long the installation takes, how often it is repeated, and whether the scaffold removal affects when the generating unit can go back into service. Once these figures are available, the setup and removal time for the permanent system can then be compared against the existing approach. Without them, a numerical claim about improved efficiency would be difficult to support.

3.3 Inspection and Maintenance

The inspection requirements are where the comparison becomes less straightforward. Temporary scaffolding has to be checked before workers use it. Ontario Regulation 213/91 covers scaffold design and inspection in Section 130 and the supervision of erection, alteration, and dismantling in Section 131 [3]. Under Section 130(1), a scaffold exceeding 15 m above its base, or 10 m for a tube-and-clamp scaffold, is also subject to additional design requirements. The University of Toronto standard also requires scaffold inspections after erection and after relocation [4]. Since the scaffold is assembled again and again, many of these activities then have to be repeated during each maintenance outage.

A permanent platform would not have the same recurring erection process. However, it would bring long-term inspection responsibilities of its own. This is particularly relevant inside a draft tube. The moisture and the operating environment mean that corrosion protection has to be considered, and the connections, supporting members, and also the guardrails would all have to stay in acceptable condition over the service life of the platform. The available publications do not explain how the McCormick platform is maintained. The

inspection interval, intended design life, protective coating system, and whether inspection responsibility sits with the station owner or with another party could not be found. Because of this, a fair comparison of the total maintenance effort is not possible from the information which is available. It would also be premature to conclude that the permanent platform needs less inspection just because it does not have to be erected for every outage. Its inspections may be less frequent, but their scope and complexity still have to be determined.

4. Limitations of the Review

Several points would need to be resolved before this research could be used to support an actual project decision.

The first is the applicable legislation. McCormick is located in Québec, and the relevant Québec requirements were not reviewed. Ontario Regulation 213/91 is a useful reference for discussing scaffolding, but it cannot be assumed to apply to the McCormick installation. There is also a further distinction for an Ontario station. Depending on the type of work, the applicable requirements may fall under O. Reg. 213/91 for construction projects [3] or under Reg. 851 for industrial establishments [7]. This determination would have to be made for the actual maintenance activity which is being carried out.

The second limitation is the lack of measured operating data. None of the sources gives a complete comparison of scaffold labour hours, permanent platform setup time, project costs, or the incidents which are associated with each arrangement. In addition, there is a concern about the independence of the information. Hatch and ZARGES are both describing solutions which are connected to their own businesses. Their publications help to explain the applications, but independent records would give much stronger support for any claim about performance.

Finally, the provisions of O. Reg. 213/91 were reviewed using the IHSA's reproduction of the regulation. The cited sections and the current wording should be confirmed through Ontario's e-Laws before they are relied on in a formal engineering assessment.

5. Conclusion

The McCormick project shows how a permanent modular platform can be used to deal with repeated maintenance access inside a hydropower station. At the same time, it also brings up the question of how the access equipment gets chosen in the first place. The scaffolding itself may be fine for the work which is being done on it. The problem is that every time it is installed again, more time is used up and the workers are also exposed to the hazards which come with putting it up and taking it down. In a station where the inspections happen often, this repeated work then ends up being a large part of the maintenance which is being done.

Of the three factors which were considered, worker safety gives the clearest reason to look into a permanent platform. Reducing the need to build and remove temporary scaffolding should reduce the exposure to those particular activities, although the size of the improvement was not measured in any of the sources which were reviewed. The time which could be saved is also worth looking into. The problem is that there is not enough published data at the moment to work out what that saving would actually be. Inspection and maintenance are also still not settled. A permanent platform avoids the repeated scaffold erection, but it then has to be kept structurally sound and serviceable for its whole intended life.

Taking all of this into consideration, selecting one system over the other based only on the available publications would not be recommended. Before doing this, a record of the current scaffold labour requirements, the frequency of maintenance access, and the expected service life and inspection needs of the proposed platform would be needed. These details would turn the comparison from a general argument in favour of permanent access into something which could actually support an engineering decision.

6. Questions for Senior Engineering Review

1. When turbine maintenance has to be repeated, is getting rid of the scaffold erection and dismantling enough on its own to justify a permanent platform from a safety point of view, or would measured labour hours and outage data normally be needed first before that decision is made?
2. Which legislation and which site requirements would then apply to the access system? In particular, how is it normally decided whether the work counts as construction or as ongoing industrial maintenance at a station like this one?
3. For a permanent platform which has been installed inside a draft tube, who is normally responsible for inspecting it over its service life (the station owner, the designer, or someone else), and how often and to what level of detail would that inspection then be expected to be done?

References

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